AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Cell structure

Every organism is built from cells, and inside each cell different structures carry out different jobs. This page covers the ultrastructure of eukaryotic and prokaryotic cells, the route a secreted protein takes through the cell, how cells are grouped into tissues and organs, and how microscopes and simple calculations let us measure what we see.

What all cells have in common

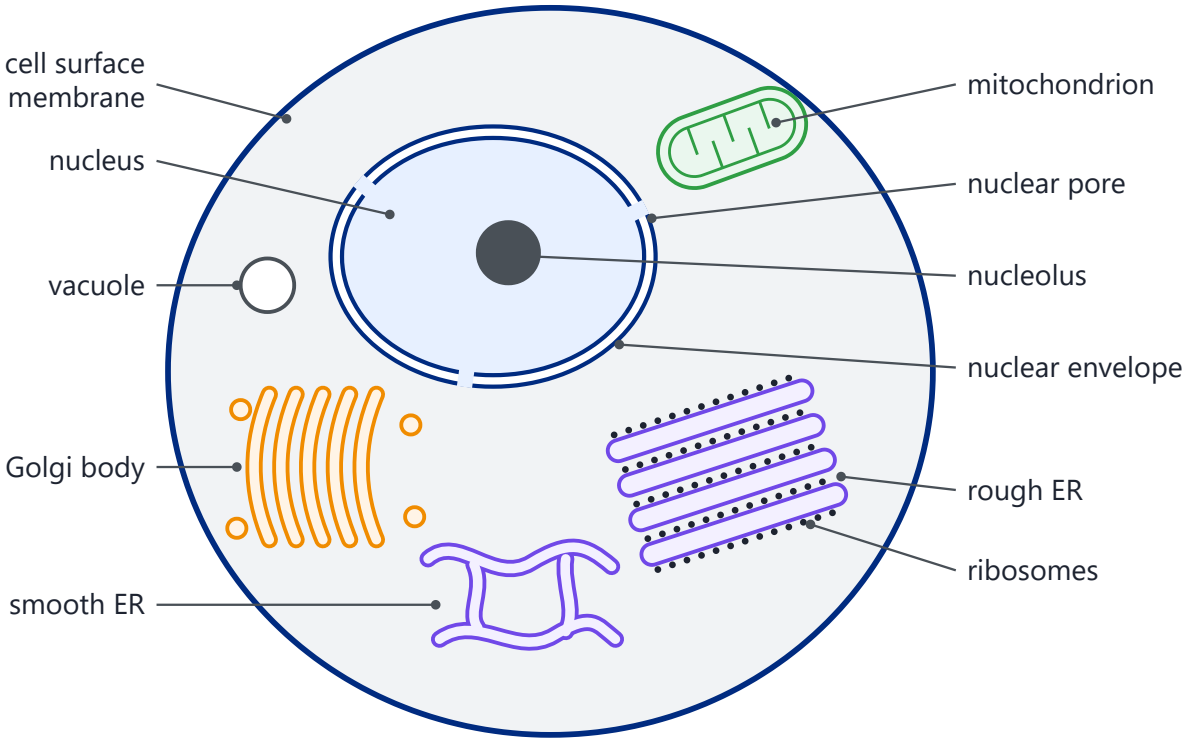
Cell theory says that the cell is the basic unit of life and that new cells only come from existing cells. Whatever the organism, its cells share a small set of features, because these are the minimum needed to stay alive and make copies of itself:

- **Cell surface membrane:** a phospholipid bilayer with proteins that separates the inside of the cell from its surroundings and controls what enters and leaves.
- **Cytoplasm:** the watery contents of the cell where many metabolic reactions take place.
- **DNA:** the genetic material, which carries the instructions for making proteins.
- **Ribosomes:** the sites where those instructions are read and proteins are assembled.

Cells then fall into two groups. **Eukaryotic** cells (animals, plants, fungi and protoctists) keep their DNA inside a nucleus and have many membrane-bound organelles. **Prokaryotic** cells (bacteria) have no nucleus and no membrane-bound organelles. Eukaryotic cells are typically 10 to 100 µm across, while prokaryotic cells are usually 0.5 to 5 µm long.

The ultrastructure of a eukaryotic cell

Ultrastructure means the detail of a cell that only an electron microscope can reveal. Most organelles are surrounded by membranes, which divide the cell into compartments. Compartments matter because each one can hold its own enzymes and conditions, so reactions that would interfere with each other are kept apart, and harmful enzymes, such as those in lysosomes, are kept away from the rest of the cell.



An animal cell as it appears in an electron micrograph. Note the double nuclear membrane with pores, ribosome-studded rough ER, and the stacked sacs of the Golgi body.

Organelle	Structure	Function
Nucleus	The largest organelle. Surrounded by a nuclear envelope of two membranes, perforated by nuclear pores. Contains chromatin: linear DNA wound around histone proteins.	Stores the genetic information and controls the cell by deciding which genes are transcribed. Pores let mRNA and ribosome subunits out into the cytoplasm.
Nucleolus	A dense, darkly staining region inside the nucleus. It has no membrane.	Makes ribosomal RNA (rRNA) and assembles ribosome subunits.
Ribosomes	Tiny particles of rRNA and protein, made of a large and a small subunit. No membrane. Eukaryotic ribosomes are 80S.	Site of translation, where amino acids are joined by peptide bonds into polypeptides. Free ribosomes make proteins used inside the cell; ribosomes on the RER make proteins for export or for membranes.
Rough endoplasmic reticulum (RER)	A system of flattened, membrane-bound sacs (cisternae) continuous with the nuclear envelope, with ribosomes on the outer surface.	Polypeptides made by its ribosomes enter the lumen, where they fold into their 3D shape. The proteins are then sent on to the Golgi apparatus in transport vesicles.
Smooth endoplasmic reticulum (SER)	A network of membrane-bound tubules with no ribosomes.	Makes and processes lipids, including phospholipids and steroids. Stores calcium ions, especially in muscle cells.

Organelle	Structure	Function
Golgi apparatus	A stack of curved, flattened cisternae surrounded by small vesicles. Vesicles from the ER fuse with one face (the cis face) and new vesicles bud off the other (the trans face).	Modifies proteins, for example by adding carbohydrate to make glycoproteins or lipid to make lipoproteins. Sorts and packages them into secretory vesicles. Also makes lysosomes.
Mitochondrion	Rod-shaped, about 1 μm wide. Two membranes: the inner one is folded into cristae. The inside, the matrix, contains enzymes, small circular DNA and 70S ribosomes.	Site of the aerobic stages of respiration, producing most of the cell's ATP. Cristae give a large surface area for the proteins of oxidative phosphorylation.
Lysosome	A small spherical vesicle bounded by a single membrane, containing hydrolytic (digestive) enzymes.	Digests material taken into the cell, breaks down worn-out organelles (autophagy), can destroy the whole cell (autolysis), and can release enzymes outside the cell by exocytosis.
Centrioles	A pair of hollow cylinders lying at right angles, close to the nucleus, each made of nine bundles of microtubules. Found in animal cells, not in most plant cells.	Before division they move to opposite poles and organise the spindle fibres that pull chromosomes apart.
Cell surface membrane	A phospholipid bilayer with proteins, glycoproteins and cholesterol (fluid mosaic model). About 7 nm thick.	Partially permeable barrier that controls the movement of substances in and out. Carries receptors and antigens for cell signalling and recognition.

KEY IDEA

Structure fits function. The cristae of a mitochondrion pack a huge area of membrane into a small space for ATP synthesis. The RER is studded with ribosomes because it receives proteins as they are made. A cell that secretes a lot of protein, such as a pancreas cell making digestive enzymes, has large amounts of RER and a well-developed Golgi apparatus.

Recognising organelles in electron micrographs

In the exam you may be shown a black and white electron micrograph and asked to name a structure. Look for membranes and shapes rather than colour:

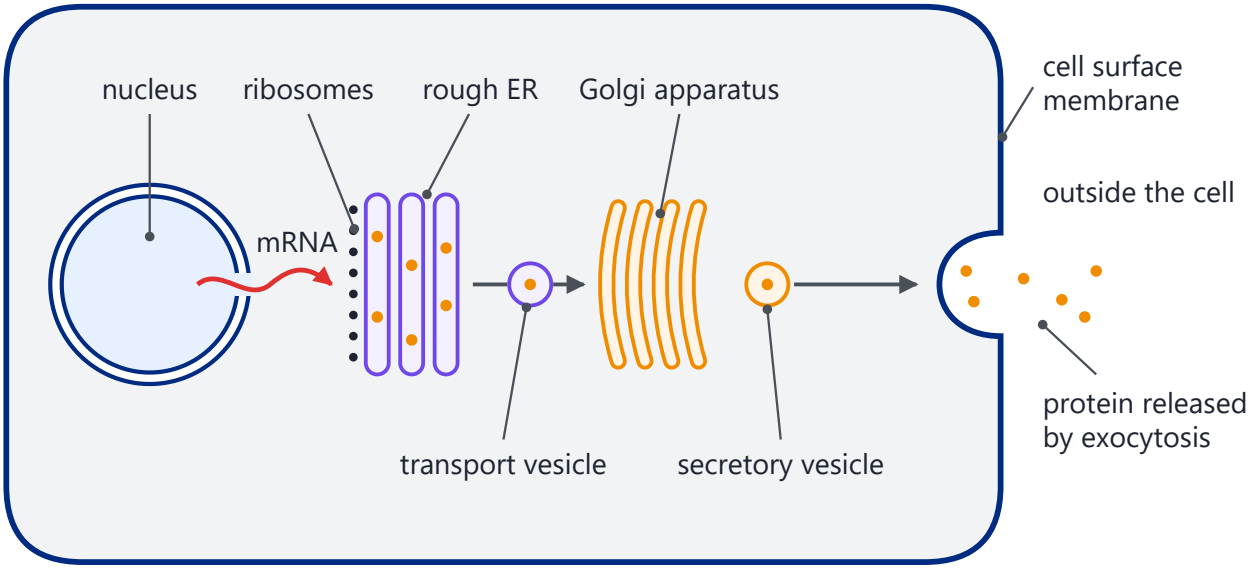
- **Nucleus:** large, round, with a double outline broken by pores and a dark patch (the nucleolus) inside.
- **Mitochondrion:** oval or sausage shaped, with a smooth double outline and inner folds running across it.
- **Golgi apparatus:** a curved stack of parallel, flattened sacs with small round vesicles gathered at the edges.
- **RER:** parallel sheets of membrane lined with tiny dark dots (ribosomes). SER looks similar but has no dots and is more tubular.
- **Lysosome:** a small, round, dark-staining body with a single membrane.
- **Centriole:** seen in cross-section as a ring of nine groups of tubes, or along its length as a short striped cylinder.

COMMON MISTAKE

Students often confuse the Golgi apparatus with a mitochondrion. A mitochondrion has two membranes, cristae, its own ribosomes and circular DNA, and is a single closed body. The Golgi apparatus is a stack of separate sacs, each with one membrane, and is surrounded by vesicles.

How proteins are made and secreted

Many cells release proteins to work outside them. Examples are digestive enzymes (extracellular enzymes) from the pancreas, antibodies from plasma cells and peptide hormones such as insulin. These proteins pass through a fixed route of organelles. Each organelle adds something, and the protein is kept inside membranes the whole time so that it never mixes with the cytoplasm.



The secretory pathway, from transcription in the nucleus to release of the finished protein at the cell surface.

The route of a secreted protein

1. In the nucleus, the gene for the protein is transcribed into mRNA, which leaves through a nuclear pore.
2. The mRNA attaches to a ribosome on the rough ER. As the polypeptide is made, it is fed into the lumen of the RER.

Making it straight into the RER keeps the protein separate from the cytoplasm, so it can be packaged for export.

3. Inside the RER the polypeptide folds into its secondary and tertiary structure, and it moves through the ER sacs.
4. Small transport vesicles pinch off the RER, carrying the protein. They move to the Golgi apparatus and fuse with its cis face.
5. In the Golgi apparatus the protein is modified, for example by adding carbohydrate to form a glycoprotein, and then sorted.
6. Secretory vesicles bud from the trans face of the Golgi and move along the cytoskeleton to the cell surface membrane.
7. The vesicle membrane fuses with the cell surface membrane and the protein is released outside the cell. This is exocytosis.

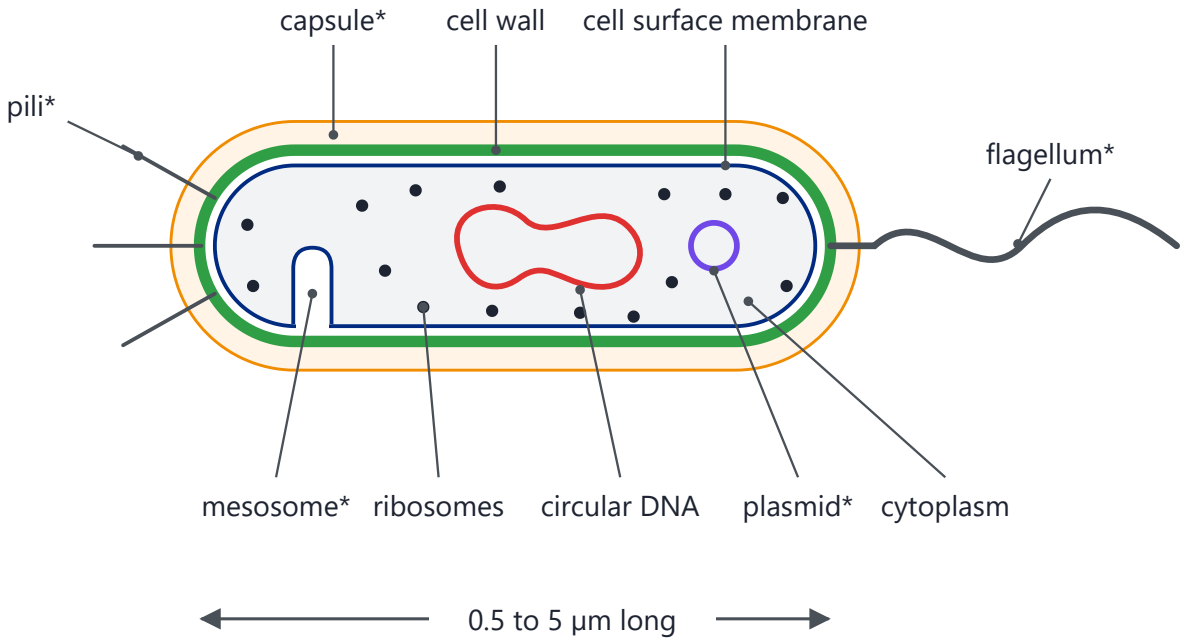
Exocytosis needs ATP, which is why secretory cells contain many mitochondria.

EXAM TIP

For full marks, name the organelles in order (ribosome on RER, RER, vesicle, Golgi apparatus, vesicle, cell surface membrane) and say what happens at each: synthesis, folding, transport, modification, packaging, exocytosis.

The ultrastructure of a prokaryotic cell

Bacteria are prokaryotes. They are far smaller than eukaryotic cells and lack a nucleus and membrane-bound organelles, but they carry out the same basic processes. Some of their structures are always present; others, marked with an asterisk in the diagram, are found only in some species.



A generalised bacterial cell. Structures with an asterisk are not present in every bacterium.

- **Circular DNA:** one large loop of DNA lying free in the cytoplasm in a region called the nucleoid. It is not wrapped around histones and is not enclosed by a membrane. It carries the genes the cell needs to survive.
- **Plasmids:** small extra rings of DNA that replicate on their own. They carry a few genes that can be useful, such as antibiotic resistance, and can be passed between bacteria.
- **Ribosomes:** smaller than eukaryotic ones (70S rather than 80S). They make proteins. Because they differ from our ribosomes, some antibiotics can target them without harming human cells.
- **Cell wall:** a rigid layer made of peptidoglycan (murein), a polymer of sugars cross-linked by short peptide chains. It keeps the cell's shape and stops it bursting when water enters by osmosis.
- **Capsule (slime layer):** a sticky outer layer in some bacteria. It protects against drying out and helps the cell avoid being engulfed by white blood cells. It also helps bacteria stick to surfaces.
- **Flagellum:** a long, whip-like protein structure that rotates from its base to push the cell through liquid.
- **Pili:** short, thin protein tubes on the surface. They attach the bacterium to surfaces or host cells, and special pili join two bacteria so that plasmids can be transferred.
- **Mesosomes:** infoldings of the cell surface membrane seen in electron micrographs of some bacteria. In exam answers they are described as increasing the membrane area for the enzymes of respiration. Many biologists now think they form when cells are prepared for electron microscopy.

Feature	Prokaryotic cell	Eukaryotic cell
Typical size	0.5–5 μm	10–100 μm
Nucleus	Absent; DNA in a nucleoid region	Present, with a double membrane
DNA	Circular, not bound to histones; plasmids often present	Linear chromosomes wound around histones
Ribosomes	70S (smaller)	80S in the cytoplasm (70S inside mitochondria and chloroplasts)

Feature	Prokaryotic cell	Eukaryotic cell
Membrane-bound organelles	None	Mitochondria, ER, Golgi, lysosomes and others
Cell wall	Peptidoglycan (murein)	Cellulose in plants, chitin in fungi, none in animals

Tissues, organs and organ systems

In a multicellular organism, cells do not all do the same job. They specialise, and similar cells work together. This division of labour means each job is done efficiently, but it also means the cells depend on each other.

- **Tissue:** a group of similar cells, with a common origin, working together to perform a particular function. Examples: squamous epithelium lining the alveoli, cardiac muscle, xylem in a plant.
- **Organ:** a structure made of several different tissues working together to carry out one or more functions. Examples: the heart, the kidney, a leaf.
- **Organ system:** a group of organs working together to carry out a major function of the body. Examples: the circulatory system (heart and blood vessels), the digestive system.

The heart shows the idea well. It contains cardiac muscle tissue to contract, connective tissue to hold it together, nervous tissue to coordinate the beat, and epithelium lining the chambers. Together these tissues make one organ that pumps blood, and the heart with the blood vessels forms the circulatory system.

EXAM TIP

If asked how a tissue differs from an organ: a tissue is made of one main type of cell doing one job, while an organ contains several tissue types and may carry out several functions. They are alike in that both consist of cells working together.

Microscopy

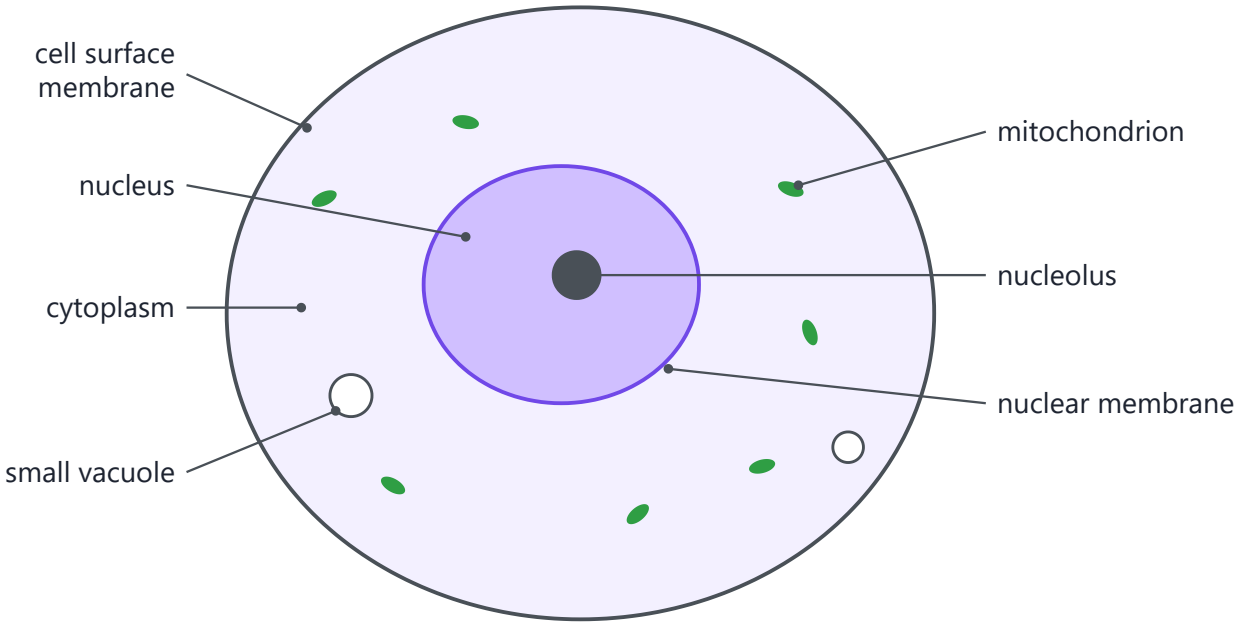
Two quantities describe what a microscope can do, and they are not the same thing.

- **Magnification:** how many times larger the image is than the real object.
- **Resolution:** the smallest distance between two points that can still be seen as two separate points. It decides how much detail can be seen.

Resolution is limited by the wavelength of the radiation used. Visible light has a wavelength of roughly 400 to 700 nm, so a light microscope cannot separate points closer than about 200 nm. Magnifying beyond about ×1500 only makes a blurred image bigger. Electrons have a much shorter wavelength, so electron microscopes resolve structures below 1 nm apart and can usefully magnify hundreds of thousands of times. That is why membranes, ribosomes and the inside of organelles are only visible with an electron microscope.

Feature	Light microscope	Transmission EM (TEM)	Scanning EM (SEM)
What forms the image	Light passing through the specimen	Electrons passing through a very thin section	Electrons bounced off the surface

Feature	Light microscope	Transmission EM (TEM)	Scanning EM (SEM)
Resolution	About 200 nm	Below 1 nm (best of the three)	A few nm
Useful magnification	Up to about ×1500	Over ×500 000	Up to about ×100 000 or more
Image	Colour (stained), 2D	Black and white, 2D section showing internal detail	Black and white, 3D view of the surface
Specimen	Can be living	Dead: fixed, stained with heavy metals, in a vacuum	Dead: coated in metal, in a vacuum



The same kind of cell seen with a light microscope after staining. The nucleus shows clearly, but mitochondria are only small dots and the ER, Golgi and ribosomes cannot be made out.

COMMON MISTAKE

A higher magnification does not show more detail if the resolution stays the same. Ribosomes (about 20 to 30 nm across) are invisible in a light microscope however much the image is enlarged, because they are smaller than its resolution limit.

Units and the magnification equation

Cells are measured in micrometres (μm) and organelles in nanometres (nm). 1 mm = 1000 μm and 1 μm = 1000 nm, so 1 mm = 1 000 000 nm. The equation linking image size, actual size and magnification is:

magnification = image size ÷ actual size

It rearranges to give actual size = image size ÷ magnification, or image size = actual size × magnification. Both lengths must be in the same unit before you divide.

Worked example: actual size from a micrograph

1. A mitochondrion measures 36 mm long on an electron micrograph with a magnification of ×12 000. Find its real length in µm.
2. Convert the image length to µm: 36 mm × 1000 = 36 000 µm.

   Converting first avoids answers that are out by a factor of 1000.
3. Actual size = image size ÷ magnification = 36 000 ÷ 12 000 = 3 µm.
4. Check: 3 µm is a sensible length for a mitochondrion (usually 1 to 10 µm).

Worked example: magnification from a scale bar

1. A scale bar labelled 5 µm is 25 mm long on the printed image.
2. Convert the bar length: 25 mm = 25 000 µm.
3. Magnification = 25 000 ÷ 5 = ×5000.

   A scale bar is just an object of known size, so the same equation applies.

Measuring with an eyepiece graticule

An eyepiece graticule is a small scale inside the eyepiece. Its divisions have no fixed size, because the image of the specimen gets larger with each objective lens while the graticule stays the same. It is calibrated against a stage micrometer, a slide engraved with a scale of known length (often 1 mm divided into 100 divisions of 10 µm).

Calibrating a graticule

1. Line up the eyepiece graticule with the stage micrometer at the objective you will use.
2. Find a point where the scales match: for example, 40 eyepiece divisions cover 10 stage divisions, which is 10 × 10 µm = 100 µm.
3. One eyepiece division = 100 µm ÷ 40 = 2.5 µm at this magnification.

4. Remove the micrometer, put the specimen on the stage and measure it. A cell 24 divisions long is 24 × 2.5 = 60 µm long.

If you change objective, the calibration must be repeated, because each division now covers a different real length.

Quick checks

QUICK CHECK

Which organelle adds carbohydrate to proteins to form glycoproteins before they are secreted?

- Rough endoplasmic reticulum
- Golgi apparatus
- Lysosome
- Nucleolus

The Golgi apparatus modifies proteins, including adding carbohydrate, and packages them into secretory vesicles. The RER is the commonest wrong answer: it receives and folds the polypeptide, but the main modification and packaging happen in the Golgi.

QUICK CHECK

Which structure is found in a prokaryotic cell but never in a eukaryotic animal cell?

- Ribosome
- Cell surface membrane
- Cytoplasm
- Peptidoglycan cell wall

Animal cells have no cell wall, and only bacteria have walls made of peptidoglycan. Ribosomes are present in both cell types; they differ in size (70S and 80S) but are not absent from eukaryotes.

QUICK CHECK

Why can an electron microscope show ribosomes when a light microscope cannot?

- Electrons have a much shorter wavelength than light, so the resolution is far higher
- An electron microscope magnifies more, and magnification alone reveals detail
- Ribosomes only exist in dead, fixed cells
- Stains used in light microscopy destroy ribosomes

Detail depends on resolution, which depends on wavelength. The shorter wavelength of electrons lets points less than 1 nm apart be distinguished. Extra magnification without better resolution only enlarges a blurred image, which is why the second option is wrong.

QUICK CHECK

A cell is 40 µm in diameter. In a drawing it is 80 mm across. What is the magnification of the drawing?

- ×2
- ×200
- ×2000
- ×20 000

80 mm = 80 000 µm, and 80 000 ÷ 40 = 2000. Forgetting to convert millimetres to micrometres gives ×2, the commonest error.

QUICK CHECK

Forty eyepiece divisions line up with 0.2 mm on a stage micrometer. How long is a cell that spans 15 eyepiece divisions?

- 5 µm
- 50 µm
- 75 µm
- 200 µm

0.2 mm = 200 µm, so one eyepiece division = 200 ÷ 40 = 5 µm. The cell is 15 × 5 = 75 µm. The answer 5 µm is the value of a single division, not the length of the cell.

QUICK CHECK

Which statement correctly distinguishes an organ from a tissue?

- An organ is made of one type of cell; a tissue contains several
- A tissue is found only in animals; organs occur in plants and animals
- An organ contains several different tissues working together; a tissue is a group of similar cells with one function
- Tissues carry out several functions; an organ carries out only one

A tissue is a group of similar cells doing a particular job, while an organ is built from several tissues. The first option reverses the two definitions, and plants have tissues too, such as xylem and phloem.

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Mitosis, meiosis and reproduction

Cells divide for two reasons: to make identical copies for growth and repair, and to make gametes for sexual reproduction. This page explains the cell cycle, what happens in each stage of mitosis, how to count dividing cells in a root tip, and how meiosis differs by halving the chromosome number and shuffling the genes.

The cell cycle

The cell cycle is the sequence of growth and division a cell goes through. Interphase is the longest part, and it is when the cell is not dividing but is doing its normal work and preparing to divide. Interphase has three phases. In G1 the cell grows and makes proteins and organelles. In S phase the DNA is copied, so each chromosome becomes two identical sister chromatids held together at the centromere. In G2 the cell checks the copied DNA and makes the proteins needed for division. Mitosis then divides the nucleus, and cytokinesis divides the cytoplasm.

The cell cycle is controlled at checkpoints. Damaged DNA or incomplete replication can stop the cycle until the problem is fixed. If a cell ignores these checkpoints, it may divide uncontrollably, which is one cause of cancer.

The stages of mitosis

Mitosis produces two nuclei that each contain the same number and type of chromosomes as the parent cell. The key to this is that each sister chromatid is pulled to opposite poles of the cell. Mitosis is a continuous process, but it is described in four stages.

- **Prophase:** chromatin condenses into visible, thick chromosomes. The nucleolus disappears and the nuclear envelope breaks down. Centrioles (in animal cells) move to opposite poles and a spindle of microtubules begins to form.
- **Metaphase:** chromosomes line up along the equator of the cell. Spindle fibres attach to the centromere of each chromosome.
- **Anaphase:** the centromeres divide and the sister chromatids are pulled apart toward opposite poles. Each chromatid is now a chromosome in its own right.
- **Telophase:** chromosomes reach the poles and decondense. A nuclear envelope forms round each set, the nucleolus reappears and the spindle breaks down.

Cytokinesis usually overlaps with late telophase. In animal cells the membrane is pinched in by a contractile ring of actin. In plant cells a new cell wall forms across the middle of the cell, starting from vesicles that fuse to make a cell plate.

KEY IDEA

Mitosis produces two genetically identical diploid cells. Identical copies are the point: a skin cell replacing a

damaged cell must have exactly the same DNA as the one it replaces.

Why mitosis matters

Mitosis is the basis of growth, because an increase in cell number increases the size of an organism. It also replaces cells that are worn out or damaged, such as skin and gut lining cells. Some organisms reproduce asexually by mitosis alone. Because the offspring are genetically identical to the parent, they inherit the same genotype, which is an advantage in a stable environment but a risk if conditions change.

Mitosis happens in meristems in plants, which are regions of stem cells such as the root tip. Meristematic cells are small, have thin walls and no large vacuole, and keep dividing throughout the plant's life.

Measuring mitosis in a root tip

A root tip squash shows mitosis under the microscope. The last 5 to 10 mm of a root tip is cut off, because this region contains the meristem. The tissue is softened in acid, which breaks down the pectin holding the cell walls together. The tip is stained with a dye such as acetocarmine, which binds to DNA and makes chromosomes visible. Pressing gently on the coverslip separates the cells into a single layer, so chromosomes can be seen clearly.

The mitotic index is the proportion of cells in a sample that are undergoing mitosis. It is calculated from the number of cells showing visible chromosomes divided by the total number of cells in the view, multiplied by 100. Because the cell cycle is mostly interphase, the mitotic index is usually low, and a higher value shows that the tissue is growing quickly.

Calculating the mitotic index

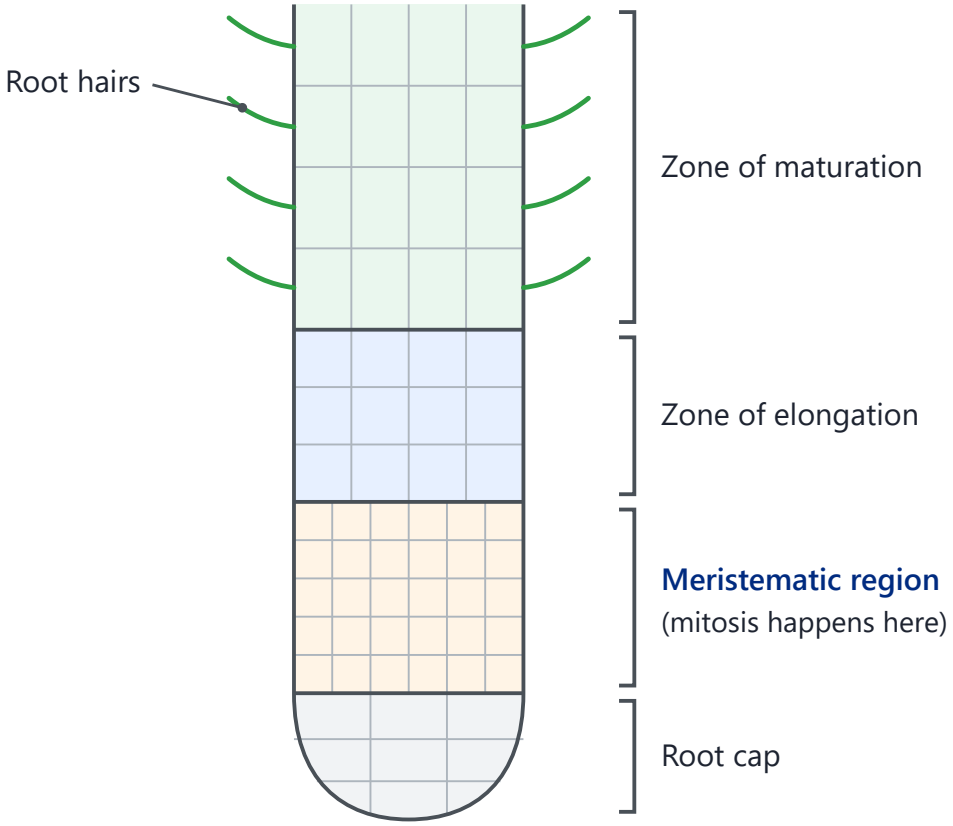
1. Count the total number of cells in one field of view.

   Use the same field and magnification for every count so the results can be compared.

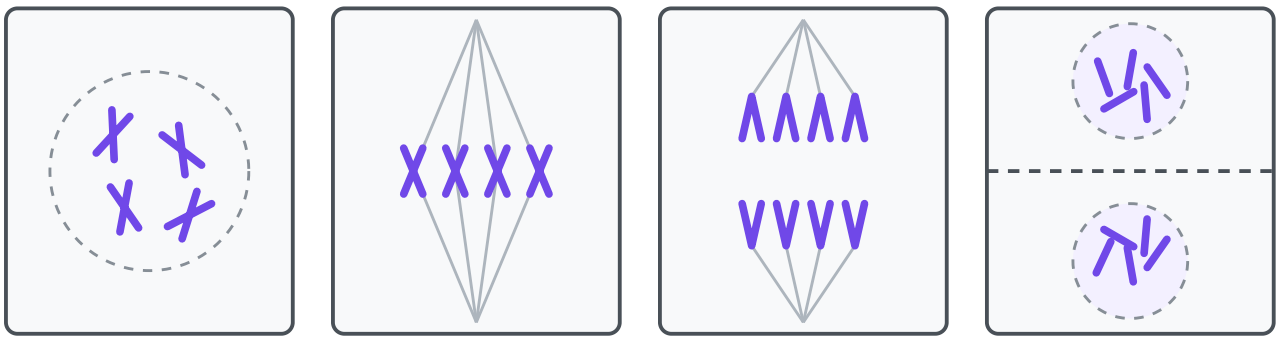
2. Count the cells that show visible chromosomes, which means cells in prophase, metaphase, anaphase or telophase.

3. Divide the number in mitosis by the total, then multiply by 100 to get a percentage.

   A percentage makes counts from different fields directly comparable.



Zones of a root tip. Mitosis happens in the meristematic region just behind the root cap.



The stages of mitosis as seen in onion root tip cells, drawn as a diagram: P (prophase), M (metaphase), A (anaphase) and T (telophase).

COMMON MISTAKE

The mitotic index does not tell you the length of each stage. A stage that lasts longer simply has more cells in it at any moment. Interphase covers most of the cell cycle, so most cells in a root tip are not dividing.

Meiosis

Meiosis is the division that produces gametes such as sperm, eggs and pollen. The purpose is to halve the chromosome number. Humans are diploid, with 46 chromosomes in 23 homologous pairs. Gametes are haploid, with 23 single chromosomes. When two gametes fuse at fertilisation, the diploid number is restored. Without meiosis, the chromosome number would double in every generation.

Meiosis involves two divisions, meiosis I and meiosis II, after one round of DNA replication. In meiosis I, homologous chromosomes pair up in prophase I, and the pairs are arranged along the equator in metaphase I. Homologous pairs are then pulled apart in anaphase I, so each daughter cell receives one chromosome from each pair. In prophase I, chromatids of homologous chromosomes can also swap sections at points called chiasmata. This is crossing over, and it creates new combinations of alleles. Meiosis II is similar to mitosis: the sister chromatids separate, and four haploid cells result.

KEY IDEA

Meiosis gives genetic variation through two mechanisms: crossing over during prophase I, and independent assortment, which is the random orientation of each homologous pair in metaphase I. Together with random fertilisation, these produce offspring that differ from their parents and from each other.

Because each homologous pair lines up independently of the others, n pairs give 2^n possible combinations of chromosomes in the gametes.

Comparing mitosis and meiosis

Feature	Mitosis	Meiosis
Number of divisions	One	Two
Daughter cells produced	Two	Four
Chromosome number of daughters	Same as parent (diploid)	Half the parent (haploid)
Genetic make-up of daughters	Identical to parent	Genetically different from parent and each other
Pairing of homologous chromosomes	No	Yes, in prophase I
Role	Growth, repair, asexual reproduction	Gamete production for sexual reproduction

Sexual and asexual reproduction

Asexual reproduction involves one parent and produces offspring by mitosis, so the offspring are clones. It is quick and needs no mate, which is useful when a population is well adapted to its habitat. Sexual reproduction uses gametes from two parents, which are fused at fertilisation. Each offspring inherits a new mix of alleles. This variation allows a population to adapt when the environment changes, such as when a new disease appears. The cost is that sexual reproduction takes more time and energy to find a mate.

EXAM TIP

In an exam answer, link each process to its result: mitosis gives identical cells because the chromosomes are copied and separated without pairing; meiosis gives variation because homologous pairs separate and crossing over mixes alleles.

Quick checks

QUICK CHECK

In which stage of mitosis do sister chromatids separate and move to opposite poles?

- Prophase
- Metaphase
- Anaphase
- Telophase

Anaphase is when the centromeres divide and the chromatids are pulled apart. Metaphase is when chromosomes line up, and telophase is when they arrive at the poles and decondense, so both are common wrong answers.

QUICK CHECK

A human cell has 46 chromosomes before meiosis. How many chromosomes does each gamete contain?

- 46
- 92
- 23
- 12

Meiosis halves the chromosome number, so each gamete has 23 single chromosomes. The answer 46 is the common mistake of forgetting that meiosis reduces the number. The answer 92 is the number of chromatids present after DNA replication, before division, so it is a trap for students who mix up chromosomes and chromatids.

QUICK CHECK

Which process produces genetic variation during meiosis I?

- DNA replication in interphase
- Anaphase of mitosis
- Crossing over between homologous chromosomes in prophase I
- Cytokinesis in plant cells

Crossing over exchanges sections between homologous chromatids in prophase I, creating new allele combinations. DNA replication copies DNA without changing it, and the other two options produce identical cells, so they are the wrong answers.

QUICK CHECK

A root tip has 200 cells in view, and 30 of them show visible chromosomes. What is the mitotic index?

- 3.0
- 15
- 15.0 %
- 30

Divide 30 by 200 and multiply by 100, which gives 15.0 %. Leaving out the multiplication by 100 gives 0.15, which is the common error.

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Development of organisms

A complex organism starts as a single cell and grows by dividing and specialising. This topic looks at stem cells, the range of cells they can give rise to, and the sources of stem cells that medicine can use, along with the ethical questions those sources raise.

1. What a stem cell is

A stem cell is an undifferentiated cell. It has not yet taken on the particular structure and job of a mature cell, but it can give rise to other cell types. It has two key properties. It can divide and renew itself, so that the stem cell population is maintained, and it can develop into one or more specialised cell types. When a stem cell divides, one daughter may remain a stem cell while the other differentiates into a specialised cell. This is how the body keeps a supply of cells to replace worn-out or damaged tissue.

KEY IDEA

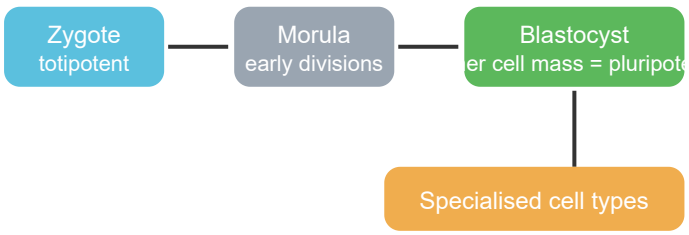
A stem cell is defined by two things: it is undifferentiated, and it can both renew itself and produce specialised cells.

2. Potency: totipotent and pluripotent cells

Potency describes how wide a range of cell types a stem cell can form. The more types it can form, the higher its potency.

A totipotent cell can form every cell type in an organism, including the cells of the placenta and umbilical cord, which lie outside the embryo itself. The zygote, made by the fusion of gametes, is totipotent, and so are the cells produced by the first few divisions of the fertilised egg.

A pluripotent cell can form almost all the cell types of the body, but not the extra-embryonic tissues such as the placenta. The inner cell mass of the blastocyst is pluripotent. Its cells are descended from the totipotent cells of the earlier divisions.



Property	Totipotent	Pluripotent
Where found	Zygote and first few divisions	Inner cell mass of the blastocyst
Cell types formed	All, including placenta and umbilical cord	Almost all embryonic and adult cell types, but not placenta
Relationship	Ancestor of pluripotent cells	Descended from totipotent cells

3. Early development: morula and blastocyst

After fertilisation the zygote divides by mitosis several times without growing in overall size. The result is a morula, a solid ball of cells. Continued division then produces a blastocyst. This is a hollow ball of cells with a fluid-filled cavity. The outer layer, the trophoblast, will form the placenta, and the inner cell mass sits at one end. It is this inner cell mass that is the source of pluripotent embryonic stem cells.

4. How stem cells give rise to specialised cells

Differentiation is the process by which a cell becomes specialised. The cell switches on some of its genes and switches off others, so it makes only the proteins that suit its job. A muscle cell makes large numbers of contractile proteins, and a nerve cell develops long extensions and the machinery to send electrical signals. Each cell still carries the full genetic information, so its specialisation comes from which genes are active rather than from losing genes.

As cells become more specialised they lose their ability to become other types. A pluripotent stem cell can form many cell types, but a cell that has committed to becoming a liver cell cannot turn into a nerve cell under normal conditions.

How a stem cell becomes a specialised cell

1. A stem cell contains the full set of genes, like every other cell in the body. Genes that every cell needs, such as those for respiration enzymes, are active, along with the genes that keep it unspecialised.
2. A signal, such as a chemical from neighbouring cells, causes some genes to be switched on and others to be switched off. Proteins called transcription factors bind to the DNA next to the genes that are switched on.

   This is differential gene expression: different cells use different parts of the same set of genes.
3. The genes that are switched on are transcribed, producing active mRNA, which leaves the nucleus through the nuclear pores.
4. The mRNA is translated on ribosomes, so the cell makes only the proteins coded for by its active genes.

5. These proteins control the cell's processes or build its structure. A muscle cell fills with contractile proteins, and a nerve cell grows long extensions and the machinery to send electrical signals.

6. The genes that are switched off are still present but are not transcribed, so their proteins are not made. As the cell becomes more specialised, it loses the ability to become other cell types.

   This is why a liver cell cannot normally become a nerve cell, even though it still carries the genes for one.

Genes are switched on by transcription factors, proteins that bind to the DNA just in front of a gene so that it can be transcribed.

5. Sources of stem cells

Stem cells used in medicine can come from three main places. Each has its own advantages and drawbacks, and these shape which source is used for which treatment.

Embryonic stem cells

Embryonic stem cells are pluripotent and come from the inner cell mass of a blastocyst. They can be obtained from spare embryos left over from in-vitro fertilisation treatment, which are donated for research. A single cell can also be removed from an embryo at the eight-cell stage for genetic testing, and the rest of the embryo can still develop normally. Embryos can be grown until the blastocyst stage, and the inner cell mass is then cultured in the laboratory.

Their great advantage is their potency: they can form most human cell types and are easy to identify and isolate. Their main drawback is ethical. Using them destroys the embryo, and the supply of spare embryos is limited.

Adult stem cells

Adult stem cells are found in tissues of growing and mature bodies, for example in bone marrow, brain, liver and skin. They can only give rise to a limited range of cells within their own tissue. White blood cells, for example, can be formed from stem cells in bone marrow.

Adult stem cells raise fewer ethical concerns, because collecting them does not destroy an embryo. If they come from the patient themselves, the risk of rejection is very low. However, they are hard to identify and isolate, they divide poorly in culture, their potency is limited, and they can accumulate DNA changes as the person ages.

Umbilical cord blood stem cells

Blood taken from the umbilical cord and placenta after birth contains pluripotent stem cells. It is collected, the stem cells are multiplied in sterile culture, and the cord blood can be frozen and stored for later use. Its advantages are that it is more versatile than bone marrow cells, the chance of immune rejection is very small, and collection does

no harm to mother or baby. Its disadvantages are that the supply is limited, collection and storage are expensive, and a large storage facility is needed.

Source	Potency	Main advantage	Main disadvantage
Embryonic	Pluripotent	Forms most cell types and is easy to isolate	Ethically controversial; uses embryos; supply limited
Adult	Limited range within a tissue	Few ethical issues; low rejection risk for the same person	Hard to isolate; limited potency; short life in culture
Umbilical cord blood	Pluripotent	Very low rejection risk; no harm to mother or baby	Limited supply; costly to collect and store

COMMON MISTAKE

Do not assume adult stem cells are as flexible as embryonic ones. Adult stem cells are usually restricted to the tissue they come from, while embryonic and cord blood stem cells are pluripotent.

6. Using stem cells in medicine

Stem cells are used to replace cells that are lost or faulty. In Parkinson's disease, dopamine-producing nerve cells in the brain are gradually lost, and the lack of dopamine makes movement stiff and difficult. Experiments in monkeys have shown that embryonic stem cells can be coaxed into dopamine-producing neurons and transplanted, which improved symptoms. Diabetes, caused by destruction of the insulin-producing cells of the pancreas, is another target, because new cells could in principle restore insulin production. Damaged nerves in the brain and spinal cord may also be repaired using stem cells. In principle, stem cells from a patient could be grown into a tissue or an organ for transplant, avoiding a donor and reducing the risk of rejection.

KEY IDEA

Medical use depends on matching the source to the need. Pluripotency is useful when many cell types are required. A patient's own adult or cord blood stem cells are useful when rejection must be avoided.

7. Ethical decisions about stem cells

Society must weigh the benefits of stem cell research against concerns about the embryo. Regulating authorities decide how old an embryo can be when it is used for research, what counts as an acceptable source of stem cells, and how research is controlled to stop unnecessary repetition. They also ensure that cloning of humans is prevented. These judgements are made on ethical grounds, and they differ between countries.

Some people argue that an embryo has the potential to become a human being, so it should not be destroyed for research. Others argue that the potential benefit to patients justifies using spare embryos that would otherwise be discarded. Adult and umbilical cord sources avoid this objection, which is one reason they are often preferred when they are suitable.

8. Quick checks

QUICK CHECK

Which statement correctly describes a pluripotent stem cell?

- It can form every cell type, including the placenta
- It can only form cells within one tissue
- It can form almost all cell types of the body but not extra-embryonic tissue
- It has already committed to one specialised cell type

Pluripotent cells can form almost all body cell types but not the placenta or other extra-embryonic tissues, which is what separates them from totipotent cells. A cell limited to one tissue is an adult stem cell, and a committed cell is no longer a stem cell.

QUICK CHECK

Where are the pluripotent embryonic stem cells found in a blastocyst?

- In the outer trophoblast layer
- In the fluid-filled cavity
- Scattered through the placenta
- In the inner cell mass

The inner cell mass gives rise to the body of the embryo and is the source of pluripotent stem cells. The outer trophoblast forms the placenta, and the fluid cavity is not made of cells.

QUICK CHECK

Why are adult stem cells less ethically controversial than embryonic stem cells?

- They are always totipotent
- They can only be used on the same species
- Collecting them does not destroy an embryo
- They never need to be cultured

Adult stem cells are taken from tissues of a living person, so no embryo is destroyed. They are not totipotent, they can be used in other species in some research, and they usually do need culturing before use.

QUICK CHECK

Differentiation in a stem cell mainly involves which change?

- Losing its genes as it becomes specialised
- Switching to a different species of genetic code
- Switching some genes on and others off so only certain proteins are made
- Dividing faster than the stem cell next to it

A specialised cell keeps the full set of genes. Differentiation changes which genes are expressed, so the cell makes only the proteins its job needs. Division rate does not define differentiation.

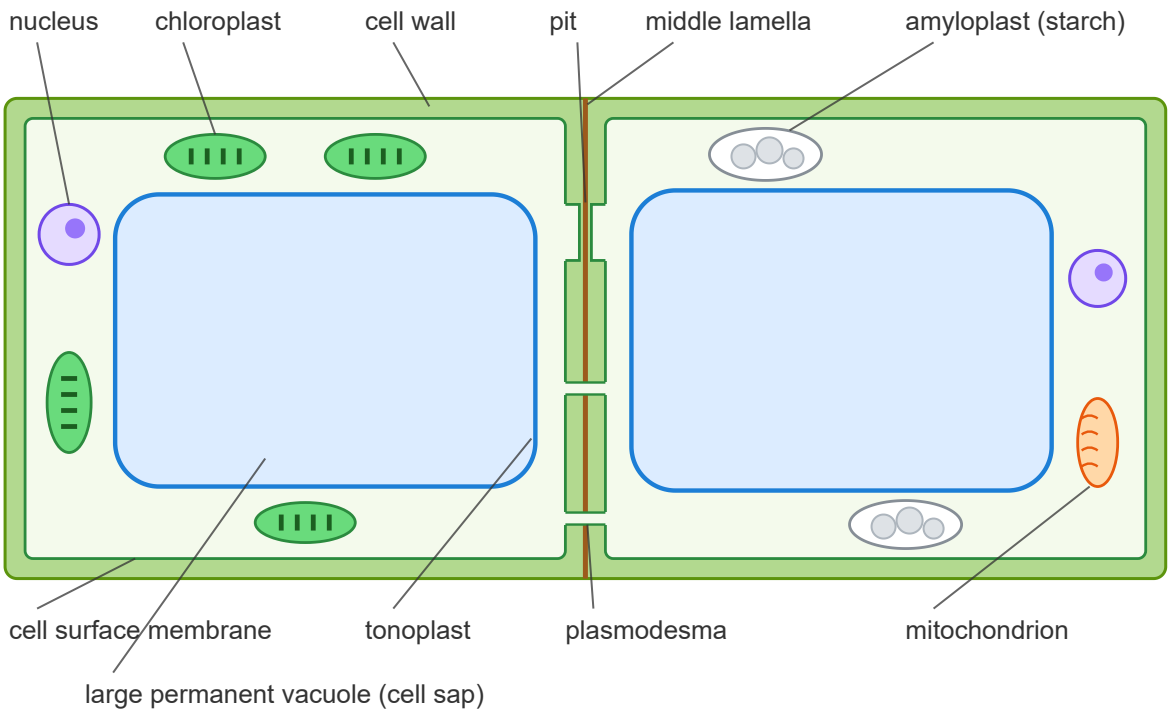
AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Plant structure and function

Plants are built differently from animals: their cells sit inside walls of cellulose, they store energy as starch, and they move water and food through specialised tissues. This page covers plant cell ultrastructure, starch and cellulose, the tissues of the stem and the fibres people make from them, the water and mineral ions plants need, and how drugs from plants are discovered and tested.

Plant cell ultrastructure

A plant cell has all the organelles of an animal cell (nucleus, ribosomes, rough and smooth ER, Golgi apparatus, mitochondria, lysosome-like vesicles), plus several structures that suit a life without moving, without a skeleton and with energy coming from light.



- **Cell wall:** a fully permeable layer outside the cell surface membrane, made mainly of cellulose microfibrils set in a matrix of other polysaccharides (hemicelluloses and pectins). It gives the cell its shape and strength, and it resists the pressure of water entering by osmosis, so the cell becomes turgid instead of bursting.
- **Middle lamella:** the thin layer between the walls of neighbouring cells, made mostly of calcium pectate. It acts like glue, holding adjacent cells together. It is the first layer laid down when a dividing cell forms a new wall.
- **Plasmodesmata** (singular plasmodesma): narrow channels through the walls of neighbouring cells, lined with cell surface membrane and filled with cytoplasm. They link the cytoplasm of one cell to the next, so water, sugars, ions and signalling molecules can pass between cells without crossing a membrane.
- **Pits:** thin areas of the wall where little or no secondary thickening is laid down. Pits in neighbouring cells line up in pairs, so water and dissolved substances can move sideways from cell to cell. They are especially

important in xylem vessels, which have thick lignified walls everywhere else.

- **Chloroplasts:** typically 2 to 10 μm long and surrounded by a double membrane (the envelope). Inside are stacks of flattened membrane sacs called thylakoids, each stack being a granum, set in a fluid stroma. The thylakoids hold chlorophyll and are the site of the light-dependent reactions; the stroma holds the enzymes of the light-independent reactions. Chloroplasts also contain starch grains, circular DNA and 70S ribosomes.
- **Amyloplasts:** colourless plastids (no chlorophyll) that make and store starch grains. They are common in storage organs such as potato tubers and seeds, where they convert sugar into insoluble starch and back again when it is needed.
- **Large permanent vacuole:** a fluid-filled sac containing cell sap, a solution of water, sugars, mineral ions, pigments and some waste products. When it is full the cytoplasm is pushed against the wall, making the cell turgid, which supports non-woody parts of the plant.
- **Tonoplast:** the partially permeable membrane surrounding the vacuole. It controls which substances move between the cytoplasm and the cell sap.

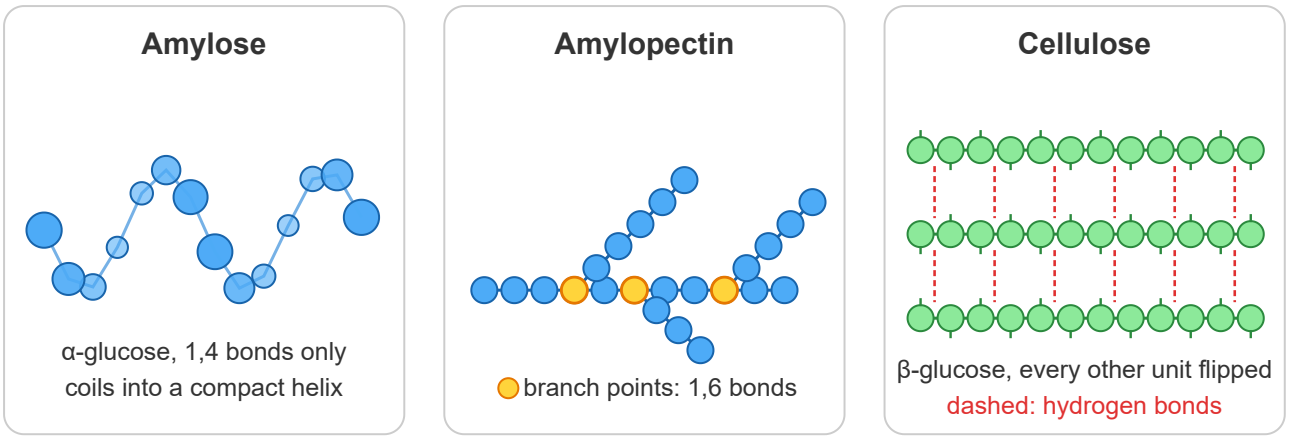
Feature	Plant cell	Animal cell
Cell wall	Present, made of cellulose	Absent
Middle lamella, plasmodesmata, pits	Present	Absent
Chloroplasts	In photosynthesising cells	Absent
Amyloplasts and starch grains	Present; starch is the energy store	Absent; glycogen granules are the energy store
Vacuole	Large and permanent, bounded by the tonoplast	Small and temporary, if present
Centrioles	Absent from most plant cells	Present
Shape	Fixed and often regular	Variable

EXAM TIP

In an electron micrograph, chloroplasts are recognised by their stacked internal membranes (grana) and double envelope, amyloplasts by large pale starch grains with little else inside, and the vacuole as a large, clear, membrane-bound space. Plasmodesmata show as fine dark lines crossing two adjacent walls.

Starch and cellulose

Starch and cellulose are both polysaccharides made of glucose monomers joined by glycosidic bonds in condensation reactions. They behave completely differently because they are built from different forms of glucose. In α-glucose the –OH group on carbon 1 points below the ring; in β-glucose it points above the ring. That single difference changes the shape of the whole polymer.



Each circle is one glucose unit. On the cellulose chains, the short ticks show the CH_2OH group pointing up or down, because every other β-glucose is turned over.

Starch: an energy store

Starch is a mixture of two polymers of α-glucose. **Amylose** is an unbranched chain with 1,4 glycosidic bonds only; the angle of the bonds makes it coil into a helix, which is held in shape by hydrogen bonds within the chain. **Amylopectin** also has 1,4 bonds, but every so often a side branch is attached by a 1,6 glycosidic bond.

- **Compact:** the coiled and branched molecules pack a lot of glucose into a small space in starch grains.
- **Insoluble:** starch does not dissolve, so it has no effect on the water potential of the cell and water does not enter by osmosis.
- **Large:** it cannot diffuse out of the cell.
- **Quick to release glucose:** the many branch ends of amylopectin give enzymes lots of places to hydrolyse glucose units off at the same time.

Cellulose: a structural molecule

In cellulose, β-glucose units are joined by 1,4 glycosidic bonds. For the –OH on carbon 1 of one unit to meet the –OH on carbon 4 of the next, every other glucose has to be turned upside down. The result is a long, straight, unbranched chain with –OH groups sticking out on both sides.

Straight chains can lie side by side. Hydrogen bonds form between the –OH groups of neighbouring chains, linking many chains into a bundle called a **microfibril**. One hydrogen bond is weak, but there are millions of them along the length of the chains, so a microfibril has a very high tensile strength. Microfibrils are wound together into larger fibres and laid down in layers in the cell wall.

Feature	Starch	Cellulose
Monomer	α-glucose	β-glucose
Glycosidic bonds	1,4 (amylose and amylopectin) and 1,6 (amylopectin branches)	1,4 only
Orientation of monomers	All the same way up	Every other one inverted
Shape	Helix (amylose) and branched chains (amylopectin)	Straight, unbranched chains

Feature	Starch	Cellulose
Hydrogen bonds	Within a chain, holding the helix	Between parallel chains, forming microfibrils
Function	Energy store	Structural support in cell walls

COMMON MISTAKE

A common error is to say cellulose is strong because of its glycosidic bonds. The strength comes from the huge number of hydrogen bonds between parallel chains, which bundle them into microfibrils. Starch chains cannot do this because they are coiled and branched.

The structure of plant cell walls

A cell wall is built in layers, from the outside in, as the cell grows and matures.

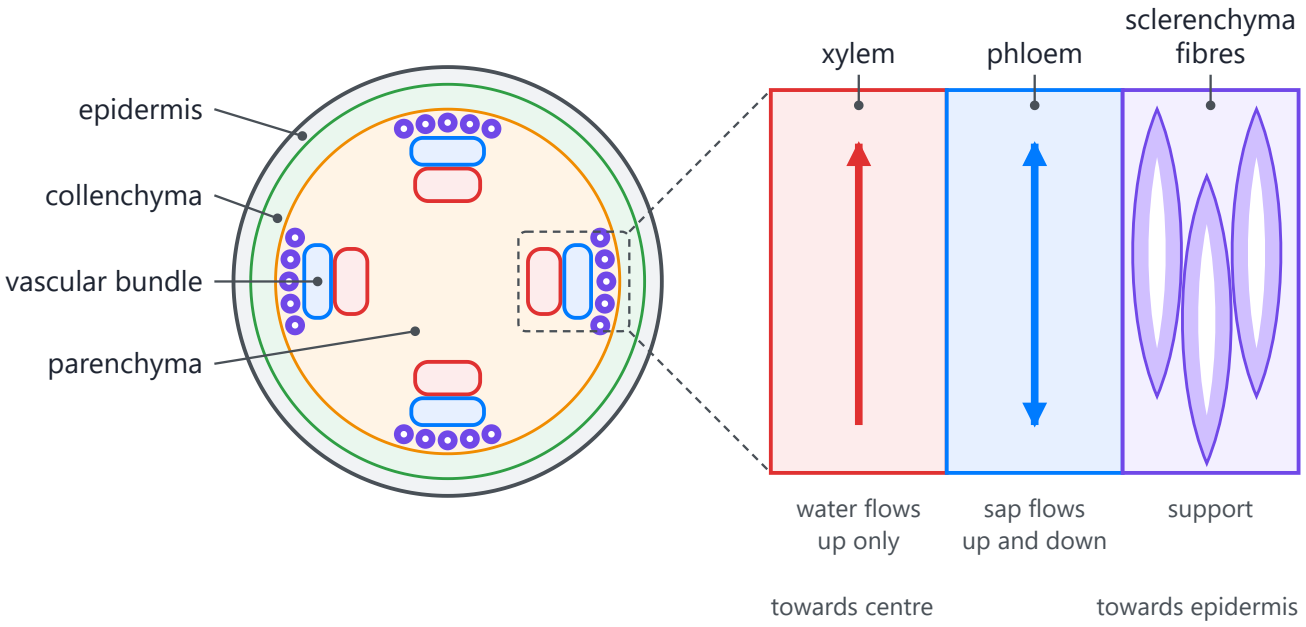
- **Middle lamella:** calcium pectate, shared between two neighbouring cells, cementing them together.
- **Primary wall:** laid down while the cell is still growing. Cellulose microfibrils run through a gel-like matrix of hemicelluloses and pectins. The wall is thin and flexible, so it can stretch as the cell expands.
- **Secondary wall:** added on the inside of the primary wall once growth has stopped, in cells that need extra strength. It is built in several layers, and in each layer the microfibrils lie parallel to each other but at a different angle from the layer next to it. Like plywood, this crossed arrangement resists forces from every direction.
- **Lignin:** a tough, waterproof polymer deposited in the walls of some cells (lignification). It fills the spaces between the microfibrils, making the wall rigid, resistant to compression and impermeable to water. Once lignified, the cell's contents die because substances can no longer pass in and out.

KEY IDEA

Secondary thickening and lignin are what make xylem vessels and sclerenchyma fibres so strong. Many layers of microfibrils at different angles give tensile strength, and lignin adds rigidity and waterproofing.

Tissues of the stem: sclerenchyma, xylem and phloem

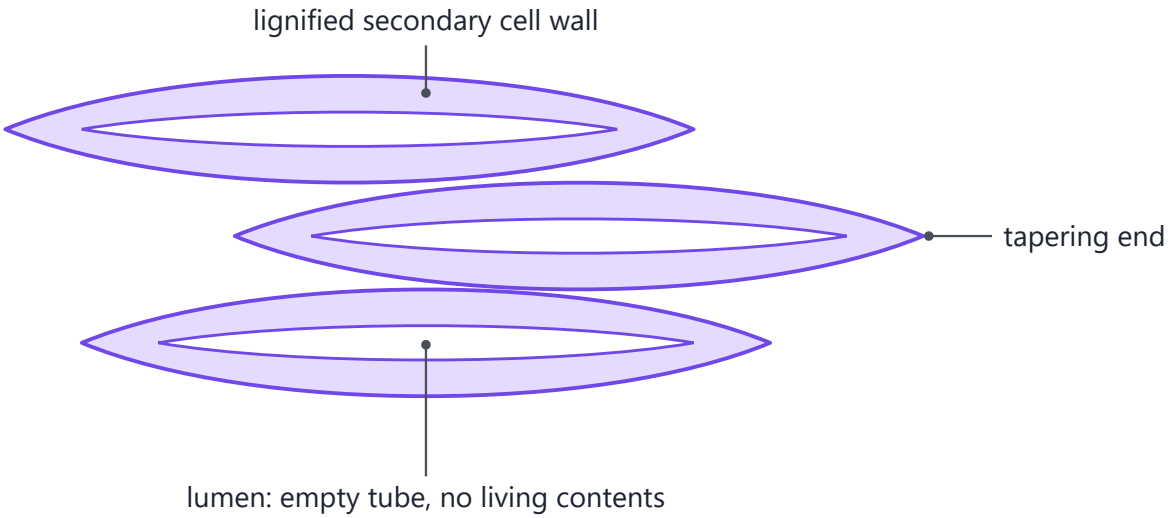
A young stem of a dicotyledonous plant is mostly parenchyma, a packing tissue of living, thin-walled cells. Running through it, in a ring near the outside, are vascular bundles. Each bundle contains xylem on the side nearest the centre, phloem on the side nearest the epidermis, and in many stems a cap of sclerenchyma fibres outside the phloem. Placing the strong tissues in a ring near the edge helps the stem resist bending, in the same way a hollow tube is stiffer than a solid rod of the same mass.



Position of the tissues in a stem, shown in transverse section (left) with one vascular bundle enlarged and seen lengthwise (right). The inner (red) part of each bundle is xylem, which carries water upwards only; the outer (blue) part is phloem, which carries sap both up and down; sclerenchyma fibres (purple) lie towards the outside.

Sclerenchyma fibres

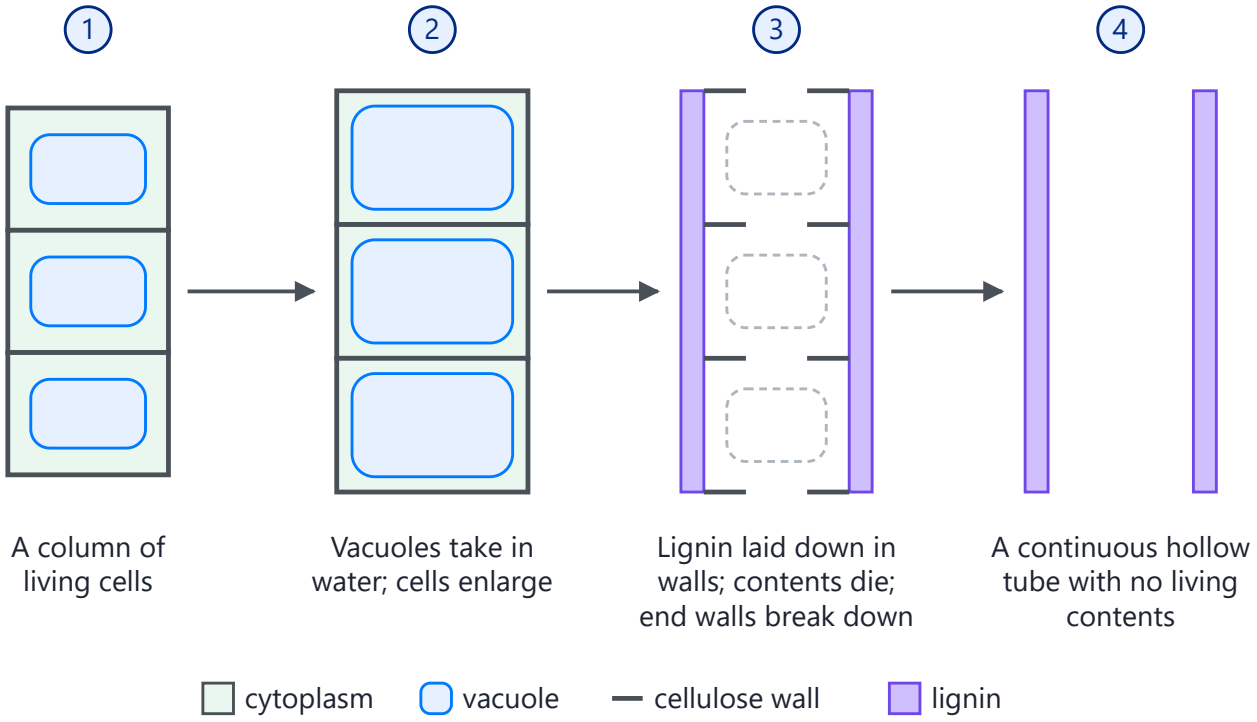
Sclerenchyma fibres are long, narrow cells with tapering, pointed ends that overlap with those of the fibres above and below. Their walls have heavy secondary thickening and are strongly lignified, so the cells are dead and hollow, with only a narrow lumen in the centre. Simple pits cross the walls. They do not carry water or food; their only job is support. Because they overlap and are bound together, a bundle of fibres is very strong when pulled, yet still flexible.



Sclerenchyma fibres: dead, hollow cells with thick lignified walls and tapering ends.

Xylem vessels

Xylem vessels develop from a column of living cells stacked end to end. Lignin is laid down in their walls, the contents die, and the end walls between the cells break down. The result is a long, continuous, hollow tube with no cytoplasm and no cross-walls.

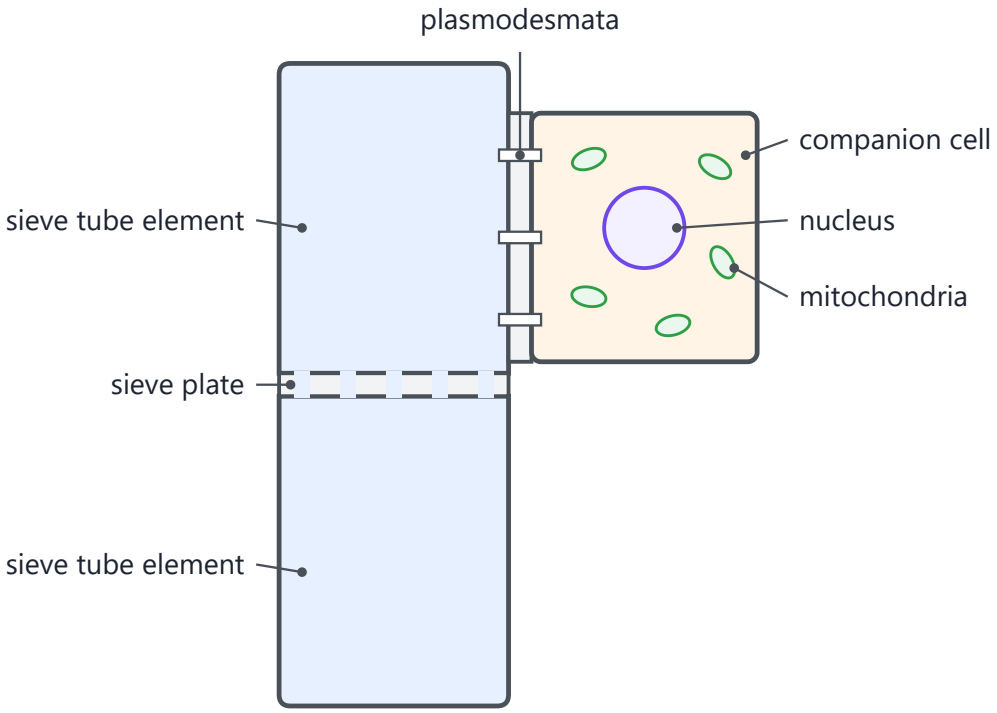


How a xylem vessel forms from a column of living cells.

- **No end walls and no contents:** water and mineral ions flow as an unbroken column with little resistance.
- **Lignified walls:** make the vessel strong enough not to collapse when the water inside is under tension, and waterproof it. In the first-formed xylem, lignin is laid down in rings or spirals, so the vessel can still stretch as the stem grows; later vessels are more fully lignified.
- **Bordered pits:** unlignified gaps that let water move sideways into neighbouring vessels or out to surrounding cells.
- **Function:** transport of water and mineral ions from the roots to the leaves in the transpiration stream, and support for the plant.

Phloem

Phloem carries organic solutes such as sucrose and amino acids from where they are made or released (sources, such as leaves) to where they are used or stored (sinks, such as roots, growing tips and fruits). This movement is called translocation, and it can go up or down the stem.



Sieve tube elements and a companion cell.

- **Sieve tube elements:** living cells joined end to end into sieve tubes. Their end walls are perforated to form sieve plates, through which the sap flows. As they mature they lose their nucleus and most organelles, leaving more room for flow.
- **Companion cells:** lie beside each sieve tube element and keep it alive. They have a nucleus, dense cytoplasm and many mitochondria, which supply the ATP used to load sucrose actively into the sieve tubes. Many plasmodesmata connect the two cells.
- **Walls:** made of cellulose and not lignified, so the cells remain alive.

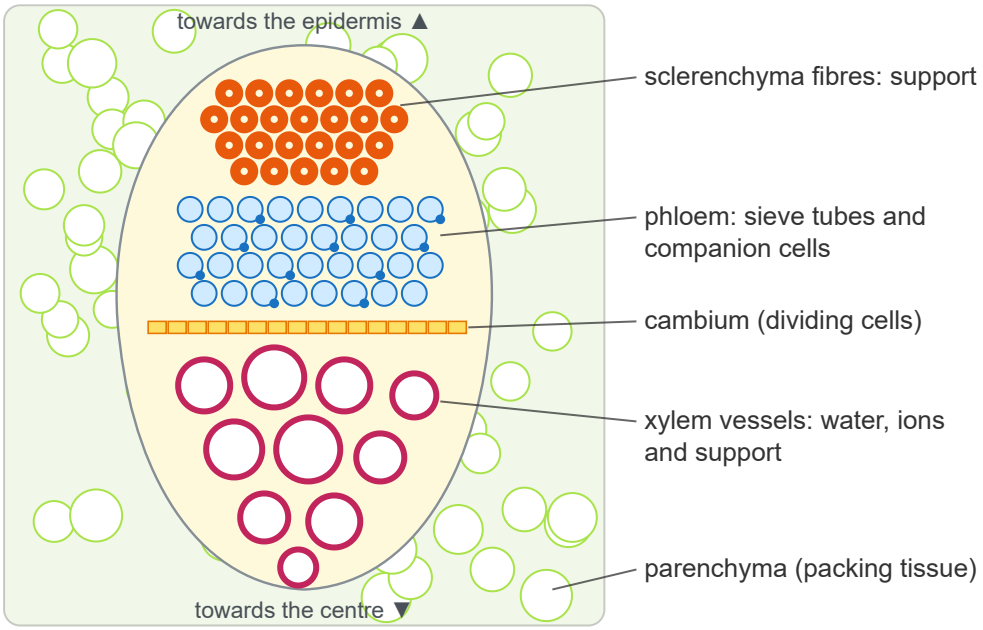
Feature	Sclerenchyma fibres	Xylem vessels	Phloem sieve tubes
Living or dead when mature	Dead	Dead	Living (with companion cells)
Wall	Thick secondary wall, heavily lignified	Lignified in rings, spirals or more completely, with pits	Cellulose only, no lignin
Ends	Tapering and closed, overlapping neighbours	Open: end walls broken down	Perforated end walls (sieve plates)
Shape	Long, narrow cells with a small lumen	Long continuous tubes with a wide lumen	Tubes of cells joined end to end
Position in a stem	Outside the phloem, towards the epidermis	Inner side of each vascular bundle	Outer side of the vascular bundle, outside the xylem
Function	Support only	Transport of water and mineral ions; support	Translocation of organic solutes such as sucrose

EXAM TIP

"Compare xylem vessels and sclerenchyma fibres" is a favourite question. Similarities: both are dead, hollow and have lignified secondary walls, and both support the plant. Differences: xylem vessels are open-ended continuous tubes with a wide lumen and transport water and ions; fibres are closed, tapering cells with a narrow lumen and do not transport anything.

Core practical: identifying the tissues in a stem

Thin transverse sections of a young stem (celery and sunflower work well) are cut with a sharp blade, stained and examined with a light microscope, then drawn.



Making and examining a stem section

1. Cut several very thin transverse sections with a sharp scalpel or razor blade, cutting away from your fingers, and keep them in water.

   Thin sections are only one or two cells thick, so light passes through and the cells do not overlap. Water stops them drying out.

2. Stain the sections, for example with toluidine blue O, which turns lignified walls a green-blue and other walls a pink-purple, or with phloroglucinol and acid, which stains lignin red.

   Staining lignin picks out the xylem and sclerenchyma, the two tissues with thick lignified walls.

3. Rinse, mount the thinnest section in a drop of water under a coverslip, and find a vascular bundle at low power.

4. Identify the tissues by their position and appearance: large empty lignified vessels on the inner side (xylem), small thin-walled cells outside them (phloem), and a group of small, very thick-walled lignified cells at the outer edge (sclerenchyma).

5. Draw a low-power plan showing the outlines of the tissues only, with no individual cells, using clear continuous lines and ruled label lines. State the magnification.

Plant fibres and their tensile strength

People have used plant fibres for thousands of years for ropes, sacks, cloth and paper. Most commercial fibres are long bundles of sclerenchyma fibres (for example flax, used for linen, and hemp, jute and sisal, used for rope and sacking), often together with xylem. They are extracted by retting: the stems are soaked so that microorganisms break down the soft surrounding tissue, leaving the tough fibres.

- **Tensile strength:** the maximum pulling force a material can withstand before it breaks, per unit cross-sectional area. Tensile strength = breaking force ÷ cross-sectional area, in N m^{-2} (Pa), usually quoted in MPa.
- **Why fibres are strong:** cellulose microfibrils held together by hydrogen bonds, laid in layers at different angles in the secondary wall, and impregnated with lignin.
- **Why fibres are flexible and light:** the cells are long and hollow, and the overlapping ends let them slide slightly rather than snap.

Core practical: determining the tensile strength of plant fibres

Measuring the breaking mass of a fibre

1. Extract fibres from the plant, for example by retting nettle or flax stems, or by pulling the vascular strands out of a celery stalk. Use fibres of the same length.

2. Clamp one end of the fibre firmly to a stand and tie a mass hanger to the other end. Place a tray of sand or a cushion underneath, and wear eye protection.

   When the fibre snaps the masses fall; the tray protects feet and bench, and goggles protect against a whipping fibre.

3. Add masses one at a time, gently and without dropping them, until the fibre breaks. Record the total mass, including the hanger, at the moment it breaks.

   Dropping a mass on would add a sudden extra force and give a falsely low value.

4. Repeat with several fibres of the same type and calculate a mean. Then repeat with other types of fibre.

   Natural fibres vary a lot, so repeats are needed to give a reliable mean and show the spread.

5. Convert to force: $F = m \times g$, with m in kg and $g = 9.81\ N\ kg^{-1}$. Measure the fibre's diameter with a micrometer or calibrated eyepiece graticule, find the area $A = \pi r^2$, and calculate tensile strength = F ÷ A.

Worked example: tensile strength

1. A fibre of diameter 0.10 mm breaks when the total mass is 400 g.
2. Force: $F = 0.400\ kg \times 9.81\ N\ kg^{-1} = 3.92\ N$.
3. Area: $r = 0.05\ mm = 5.0 \times 10^{-5}\ m$, so $A = \pi \times (5.0 \times 10^{-5})^2 = 7.85 \times 10^{-9}\ m^2$.
4. Tensile strength $= 3.92 \div 7.85 \times 10^{-9} = 5.0 \times 10^{8}\ Pa = 500\ MPa$.

   Dividing by area lets fibres of different thickness be compared fairly: a thick strand can hold more mass yet still be a weaker material.

The simulation below lets you run the practical. The breaking loads are simulated but behave like real fibres, which vary from one sample to the next.

COMMON MISTAKE

Keep the fibre length, the way the fibre is clamped and the rate of adding mass the same for every test. Knots and clamps create weak points, so a fibre that breaks at the clamp should be recorded but treated with caution.

Plant products and sustainability

A sustainable resource is one that can be replaced as fast as it is used, so it will still be available for future generations. Plant fibres and starch are renewable because more plants can be grown, whereas oil-based products depend on crude oil, which will run out.

- **Plant fibres** can replace synthetic fibres such as nylon in ropes and fabrics, and are used to reinforce plastics and building materials. They are biodegradable and take less energy to produce, and the crops absorb carbon dioxide as they grow.

- **Starch** can be made into bioplastics, such as thermoplastic starch for packaging and loose fill, or fermented to produce lactic acid for polylactic acid (PLA) plastics. Starch is also fermented to make bioethanol, a fuel.

Advantages of plant-based materials	Disadvantages
Renewable: crops can be regrown every year	Land used for these crops cannot be used to grow food, which may raise food prices
Biodegradable: broken down by bacteria and fungi, so less waste in landfill and less plastic pollution	Often more expensive to produce than oil-based plastics at present
Save crude oil for uses with no alternative	Growing crops uses fertilisers, pesticides, water and fuel for machinery
Plants absorb carbon dioxide while growing, so overall emissions can be lower	Some bioplastics need special composting conditions to break down quickly

Water and inorganic ions

Plants absorb water and mineral ions through their root hair cells. Ions are taken up mostly by active transport, against their concentration gradient, then carried to the rest of the plant in the xylem.

- **Water:** a reactant in photosynthesis; keeps cells turgid, which supports leaves and soft stems and opens stomata; the solvent in which reactions happen; the medium that carries ions in the xylem and sugars in the phloem; and evaporation from leaves cools the plant.
- **Nitrate ions (NO_3^-):** provide the nitrogen to make amino acids and so proteins, including enzymes; also nucleotides for DNA, RNA and ATP, and chlorophyll. Deficiency: stunted growth and yellowing of the older, lower leaves first.
- **Calcium ions (Ca^{2+}):** form calcium pectate in the middle lamella, which holds neighbouring cells together; also affect membrane permeability. Deficiency: poor growth of new tissue, so the growing tips die back and young leaves are pale and distorted.
- **Magnesium ions (Mg^{2+}):** part of the chlorophyll molecule, so needed for photosynthesis; also activate some enzymes. Deficiency: yellowing between the veins (chlorosis) of older leaves, then reduced growth.

EXAM TIP

Nitrate and magnesium deficiencies show first in older leaves because the plant moves these ions out of old leaves to the young growing ones. Calcium cannot be moved once it is built into cell walls, so a calcium shortage shows first at the growing tips.

Core practical: investigating plant mineral deficiencies

Growing seedlings without one ion

1. Set up identical containers of mineral solution: one complete, and others each lacking one ion (no nitrate, no calcium, no magnesium). A container of distilled water can be added for comparison.

   The complete solution is the control: any difference in the other plants must be caused by the missing ion.

2. Put seedlings of the same species, age and size into each solution, supported so the roots are in the liquid. Cover the containers with foil or black paper.

   Darkness around the roots stops algae growing in the solution and competing for the ions.

3. Keep all the plants at the same light intensity and temperature, and bubble air through the solutions or replace them regularly.

   Roots need oxygen for the aerobic respiration that powers active uptake of ions.

4. After a few weeks, record the height, number and colour of leaves, root length and dry mass of each plant, and use several plants per solution for a mean.

Drugs and antimicrobials from plants

Plants cannot run from attackers, so many have evolved chemical defences, including substances that kill or inhibit bacteria and fungi, or deter animals from eating them. Some of these substances are useful to people as medicines. Examples include aspirin, developed from salicin, a compound in willow bark that the body converts to salicylic acid, which relieves pain and fever; digoxin from foxglove, used for heart conditions; quinine from the bark of the cinchona tree, used against malaria; and morphine from the opium poppy, a strong painkiller. Extracting and purifying the active substance means patients can be given a known, consistent dose, because the amount in a plant varies with its age, the season and even the time of day.

Conditions bacteria need to grow

- **Nutrients:** a source of carbon (such as glucose) for respiration and for building molecules, a source of nitrogen (such as amino acids) for making proteins and nucleic acids, and vitamins and mineral ions. Nutrient agar or nutrient broth supplies these.
- **Water:** needed for metabolic reactions and to dissolve nutrients.
- **A suitable temperature:** enzymes work fastest near their optimum. In school laboratories cultures are incubated at no more than 25 °C, which reduces the chance of growing bacteria that are adapted to human body temperature and could cause disease.
- **A suitable pH:** most bacteria grow best close to neutral.
- **Oxygen:** needed by aerobic bacteria. Some bacteria are anaerobic and grow without it, and some can grow either way.

Core practical: investigating the antimicrobial properties of plants

Testing plant extracts on a bacterial lawn

1. Crush a known mass of plant material (for example garlic, mint or ginger) with ethanol using a pestle and mortar, then filter. Each extract should be made the same way.

   Many antimicrobial substances in plants dissolve better in ethanol than in water.

2. Using aseptic technique, spread a known volume of a bacterial culture evenly over an agar plate to make a lawn.

3. Soak sterile paper discs of the same size in each extract, and one disc in ethanol only as a control. Let the discs dry, then place them on the agar with sterile forceps, well spaced apart.

   The ethanol must evaporate, otherwise it would kill bacteria itself. The control disc shows whether any killing is due to the solvent rather than the plant.

4. Tape the lid on with two short strips (not all the way round), label the base, and incubate upside down at 25 °C for 24 to 48 hours.

   Partial taping stops the lid coming off but lets oxygen in, which discourages anaerobic bacteria, many of which are pathogens. Upside down stops condensation dripping onto the agar.

5. Measure the diameter of the clear zone around each disc, across two diameters at right angles, and calculate the area of the zone with πr^2. A larger clear zone means a more effective antimicrobial at that concentration.

KEY IDEA

Aseptic technique protects you from the bacteria and protects the culture from contamination: disinfect the bench, flame the inoculating loop until red hot and let it cool, flame the neck of culture bottles, lift the lid of a plate only a little, work close to a lit Bunsen burner, and sterilise all used cultures and equipment in an autoclave (about 121 °C for 15 minutes) before disposal.

Developing and testing new drugs

William Withering and digitalis

In 1775 the English physician William Withering heard of a herbal remedy that relieved dropsy, the swelling caused by fluid building up in the body's tissues, which is often the result of heart failure. The remedy was a mixture of

many herbs. He worked out that the active ingredient came from foxglove (*Digitalis purpurea*) and then spent about ten years testing preparations of foxglove leaves, often described as his "digitalis soup", on 163 patients.

Withering tried leaves picked at different times, fresh and dried, and found powdered dried leaves gave the most consistent effect. He started with small doses and increased them until the swelling went down or side effects such as nausea, vomiting, disturbed vision and a very slow pulse appeared, then used a slightly lower dose. Some patients were poisoned along the way. He published his careful records in 1785. The substances in foxglove, now called cardiac glycosides, strengthen the heartbeat, and digoxin is still used today.

Modern drug testing

A new drug must be shown to be safe (no unacceptable side effects), effective (it treats the condition better than nothing or than existing drugs), and given at the right dose. Modern testing follows a strict sequence.

Stages in testing a new drug

1. **Pre-clinical testing:** the drug is tested on cells and tissues grown in the laboratory, then on animals such as rodents.

   This checks the drug acts as expected in living tissue, shows how it is absorbed and broken down in a whole organism, and screens out toxic compounds before any human takes them.

2. **Phase 1:** a small number of healthy volunteers (typically 20 to 100) are given the drug, starting with very low doses.

   The aim is safety: side effects, a safe dose range, and how the body absorbs, distributes and excretes the drug.

3. **Phase 2:** a few hundred patients who have the disease are treated.

   This tests whether the drug actually works against the disease, finds the best dose and records side effects in people who are ill.

4. **Phase 3:** a large trial with thousands of patients, usually randomised and double-blind, comparing the new drug with a placebo or with the best existing treatment.

   Large numbers allow statistical tests to show whether any improvement is significant, and reveal rarer side effects.

5. If all goes well the results are reviewed by a regulatory body, which may license the drug. Its safety continues to be monitored after it is in use.

- **Placebo:** a dummy treatment that looks exactly like the drug but has no active ingredient. Patients sometimes improve simply because they believe they are being treated (the placebo effect), so the drug must be shown to do better than a placebo.
- **Double-blind trial:** neither the patients nor the doctors know who is receiving the drug and who the placebo. This removes bias in how patients report their symptoms and in how doctors assess them.
- **Randomised:** patients are allocated to each group at random, so that the groups are similar in age, sex and severity of illness.

COMMON MISTAKE

Giving a placebo to seriously ill patients would be unethical if a working treatment already exists. In that case the new drug is compared with the best existing treatment instead.

Withering's approach	Modern protocol
Tested directly on sick patients	Tested on cells, tissues and animals first, then on healthy volunteers
Crude plant extract with an uncontrolled amount of active ingredient	Pure active ingredient, so the dose is exact
Dose found by trial and error on patients, with some poisoned	Safe dose range established in phase 1 before patients are treated
No control group or placebo; patients and doctor knew the treatment	Randomised, double-blind, placebo-controlled trials
163 patients, no statistical analysis	Thousands of patients and statistical tests of significance
Similar: careful observation, recording of side effects, gradual adjustment of the dose, and publication of the results	Similar: the same principles, now required by law and checked by regulators

Quick checks

QUICK CHECK

Which structure links the cytoplasm of two neighbouring plant cells?

- Middle lamella
- Plasmodesma
- Tonoplast
- Pit

Plasmodesmata are channels filled with cytoplasm passing through both cell walls. A pit is only a thin area of wall that water can cross, and the middle lamella is the calcium pectate layer that glues the walls together, so neither connects the cytoplasm.

QUICK CHECK

Why can cellulose form microfibrils but starch cannot?

- Cellulose has 1,6 glycosidic bonds that cross-link the chains
- Starch is made of β-glucose, which cannot form hydrogen bonds
- Cellulose chains are straight, so parallel chains can be held together by many hydrogen bonds
- Cellulose chains coil into a helix that locks into neighbouring helices

Alternate β-glucose units are inverted, giving straight chains whose –OH groups form hydrogen bonds with neighbouring chains. Cellulose has only 1,4 bonds, and it is starch, not cellulose, that is made of α-glucose and coils into a helix.

QUICK CHECK

Which feature is shared by xylem vessels and sclerenchyma fibres but not by phloem sieve tubes?

- End walls perforated by sieve plates
- Transport of water and mineral ions
- Living contents supported by companion cells
- Lignified secondary cell walls

Both xylem vessels and sclerenchyma fibres have thick lignified walls and are dead. Phloem walls are cellulose only. Transporting water is a job for xylem alone, so that option is the commonest trap.

QUICK CHECK

A plant shows yellowing between the veins of its older leaves. Which ion is most likely to be lacking?

- Magnesium
- Calcium
- Nitrate
- Sodium

Magnesium is part of chlorophyll, and a shortage causes yellowing between the veins of older leaves first. Calcium shortage shows at the growing tips and young leaves. Nitrate shortage causes general yellowing of older leaves with stunted growth rather than yellowing confined to the areas between the veins.

QUICK CHECK

A fibre with a cross-sectional area of 2.0×10^{-8} m² breaks under a force of 8.0 N. What is its tensile strength?

- 1.6×10^{-7} Pa
- 4.0×10^{7} Pa
- 4.0×10^{8} Pa
- 2.5×10^{-9} Pa

Tensile strength = force ÷ area = $8.0 \div 2.0 \times 10^{-8} = 4.0 \times 10^{8}$ Pa (400 MPa). Multiplying instead of dividing gives 1.6×10^{-7} Pa, and slipping a power of ten gives 4.0×10^{7} Pa.

QUICK CHECK

Why is a phase 3 drug trial carried out double-blind?

- So that the patients do not suffer side effects
- So that neither patients nor doctors can bias the results by knowing who has the drug
- So that the drug can be compared with animal results
- So that a smaller number of patients is needed

If either the patients or the doctors know who has the real drug, their expectations can affect what is reported and recorded. Blinding does not prevent side effects, and phase 3 trials still need thousands of patients.

QUICK CHECK

In an antimicrobial investigation, why is one disc soaked in ethanol only?

- To provide nutrients for the bacteria
- To sterilise the agar around the other discs
- To make the clear zones larger
- To show that any clear zones are caused by the plant extract rather than the solvent

The ethanol-only disc is the control. If it also produced a clear zone, the solvent had not evaporated and the results could not be credited to the plant substances.

AS · Unit 2 · WBI12 Cells, Development, Biodiversity and Conservation

Classification, biodiversity and conservation

Life on Earth is enormously varied, and that variety is now under threat from human activity. This page covers how biodiversity is defined and measured, how organisms are adapted to their niches, how natural selection changes allele frequencies and produces new species, how living things are classified into three domains, and how zoos and seed banks help conserve endangered species.

Biodiversity and endemism

Biodiversity is the variety of living organisms in an area. It can be described at several levels: the variety of habitats and ecosystems, the variety of species in a habitat, and the variety of alleles within a species. The specification focuses on the last two: species diversity within a habitat, and genetic diversity within a species.

- **Species richness:** the number of different species present in a habitat. It is the simplest measure of biodiversity, but it ignores how many individuals of each species there are.
- **Genetic diversity:** the variety of alleles in the gene pool of a species or population. A species with high genetic diversity is more likely to contain individuals that can survive a change in the environment, such as a new disease.
- **Gene pool:** all the alleles of all the genes present in a population at a given time.
- **Endemism:** a species is endemic to an area if it is found naturally only in that area and nowhere else in the wild. Ring-tailed lemurs are endemic to Madagascar and koalas to Australia.

Endemic species matter for conservation because if their one habitat is destroyed, the species is lost worldwide. They often live in small, isolated populations, so they tend to have low genetic diversity as well. Islands and isolated mountain ranges have many endemic species because populations there have been cut off from others for a long time and have evolved separately. The places with the most species are not always the places with the most endemic species, which makes it hard to decide which areas to protect first.

Measuring biodiversity in a habitat

It is rarely possible to count every organism in a habitat, so a sample is taken. Plants and slow-moving animals are sampled with quadrats placed at random coordinates (generated from random number tables or a calculator), so that the sample is not biased by where the investigator chooses to look. Other methods include pitfall traps and sweep nets for insects and kick sampling in streams. The species found in each sample are identified with a key and counted.

Species richness alone can be misleading. Imagine two fields, each with five plant species. In one, all five are about equally common; in the other, one grass makes up almost everything and the other four are rare. The first is clearly more diverse. The **index of diversity** takes into account both the number of species and how evenly the individuals are spread among them:

$$D = N(N - 1) \div \Sigma n(n - 1)$$

- **N:** the total number of organisms of all species in the sample.
- **n:** the number of organisms of one particular species.
- **Σ:** "the sum of": work out n(n − 1) for every species and add the results.

The higher the value of D, the greater the biodiversity. The lowest possible value is 1, which happens when only one species is present. Because the index can be calculated in the same way anywhere, it lets habitats be compared and changes in one habitat to be followed over time.

Worked example: index of diversity

1. A sample from a rock pool contains: limpets 8, periwinkles 5, anemones 4, crabs 2, starfish 1. So N = 8 + 5 + 4 + 2 + 1 = 20.
2. Work out n(n − 1) for each species: 8 × 7 = 56, 5 × 4 = 20, 4 × 3 = 12, 2 × 1 = 2, 1 × 0 = 0.

   A species with only one individual contributes nothing to the sum, but it still counts towards N.
3. Σn(n − 1) = 56 + 20 + 12 + 2 + 0 = 90.
4. N(N − 1) = 20 × 19 = 380.
5. D = 380 ÷ 90 = 4.22.

COMMON MISTAKE

A common slip is to use N(N − 1) on the bottom and Σn(n − 1) on the top. The total for the whole sample always goes on the top, so D is never less than 1.

Measuring genetic diversity within a species

Genetic diversity can be measured directly by looking at DNA. A region of DNA that differs between individuals is cut out or copied and the fragments are separated by gel electrophoresis. If an individual is homozygous for that region, both copies of the DNA are the same length and give one band. If it is heterozygous, the two copies differ and give two bands.

The **heterozygosity index** is the proportion of individuals that are heterozygous for the gene or region studied:

H = number of heterozygotes ÷ number of individuals in the population

For example, if 9 out of 30 toads tested show two bands, H = 9 ÷ 30 = 0.30. A high value means many alleles are present and the population has plenty of genetic diversity. A low value suggests a population that has lost alleles,

for example because it is small, isolated or inbred, and it is likely to be less able to adapt if conditions change. In practice the index is averaged over many genes.

Niches and adaptations

A **niche** is the role of a species in its habitat: everything about how it lives, including what it feeds on, what feeds on it, where it lives within the habitat, when it is active and the range of conditions it can tolerate. Two species cannot occupy exactly the same niche in the same habitat for long, because they would compete for the same resources until one of them was excluded. Similar species can share a habitat by feeding on different foods, feeding at different times or using different parts of the habitat.

An **adaptation** is a feature that increases an organism's chance of surviving and reproducing in its niche. Adaptations are inherited, so successful individuals pass them on. They are grouped into three kinds:

Type	What it is	Examples
Anatomical	Structural features of the body, inside or out	Thick blubber in seals and whales to insulate against cold water; the sticky glandular hairs of a sundew that trap insects; a camel's broad, flat feet that spread its weight on sand and long eyelashes that keep sand out of its eyes
Physiological	How the body's processes work, often at the level of enzymes and chemical reactions	Diving mammals slow their heart rate under water and redirect blood to the brain and heart, and their muscles tolerate anaerobic respiration; a camel's kidneys make very concentrated urine, and its body temperature can rise by several degrees during the day, so less water is used for sweating
Behavioural	What the organism does, either inherited (instinctive) or learned	Birds migrating to avoid winter food shortages; camels sitting with their legs tucked under them early in the day, so less body surface touches the hot sand later; animals huddling together to reduce heat loss

EXAM TIP

In an exam, name the type of adaptation and explain how it helps survival or reproduction. "Thick fur" alone earns little; "thick fur traps a layer of air, reducing heat loss by conduction, so less energy is used to keep warm" earns the marks.

Natural selection and allele frequencies

Within any population, individuals differ. New alleles arise by mutation, and meiosis and random fertilisation shuffle existing alleles into new combinations. The proportion of all the copies of a gene in a gene pool that are one particular allele is the **allele frequency**. Natural selection changes allele frequencies, and this change over generations is evolution.

How natural selection leads to adaptation

1. There is variation within the population, caused by different alleles. New alleles arise by mutation.
2. A selection pressure acts on the population, for example a predator, a disease, a shortage of food or a change in climate.
3. Individuals with an allele that gives an advantage under this pressure are more likely to survive and reproduce. Individuals without it are more likely to die before reproducing.
4. The survivors pass the advantageous allele to their offspring.
5. Over many generations the frequency of the advantageous allele in the gene pool increases, and the population becomes better adapted to its environment.

Individuals do not change during their lifetime; it is the proportion of individuals carrying the allele that changes.

COMMON MISTAKE

Avoid writing that organisms "adapt because they need to" or that a mutation happens "in order to" help survival. Mutations are random. Selection only acts on variation that already exists.

The Hardy-Weinberg equation

The Hardy-Weinberg principle describes a population that is not evolving: its allele and genotype frequencies stay the same from one generation to the next. For a gene with two alleles, A (dominant, frequency p) and a (recessive, frequency q):

$$p + q = 1 \quad \text{and} \quad p^2 + 2pq + q^2 = 1$$

Here p^2 is the frequency of the homozygous dominant genotype (AA), $2pq$ the frequency of heterozygotes (Aa) and q^2 the frequency of the homozygous recessive genotype (aa). The equilibrium only holds if five conditions are met: the population is very large, mating is random, there are no mutations, there is no migration into or out of the population, and there is no natural selection (all genotypes are equally likely to survive and reproduce).

Real populations never meet all these conditions exactly, which is what makes the equation useful. Allele frequencies are calculated in one generation and the expected genotype frequencies compared with the frequencies observed later. If the observed frequencies differ significantly from those expected, at least one condition is not being met: the allele frequencies are changing, and the population is evolving, for example because of selection.

Worked example: Hardy-Weinberg

1. In a population of moths, 9 % are pale, the phenotype caused by the recessive genotype aa. Find the percentage of heterozygous (carrier) moths.

2. Only aa individuals show the recessive phenotype, so $q^2 = 0.09$.

   Always start from the recessive phenotype, because AA and Aa look the same and cannot be counted separately.

3. $q = \sqrt{0.09} = 0.3$, so $p = 1 - 0.3 = 0.7$.

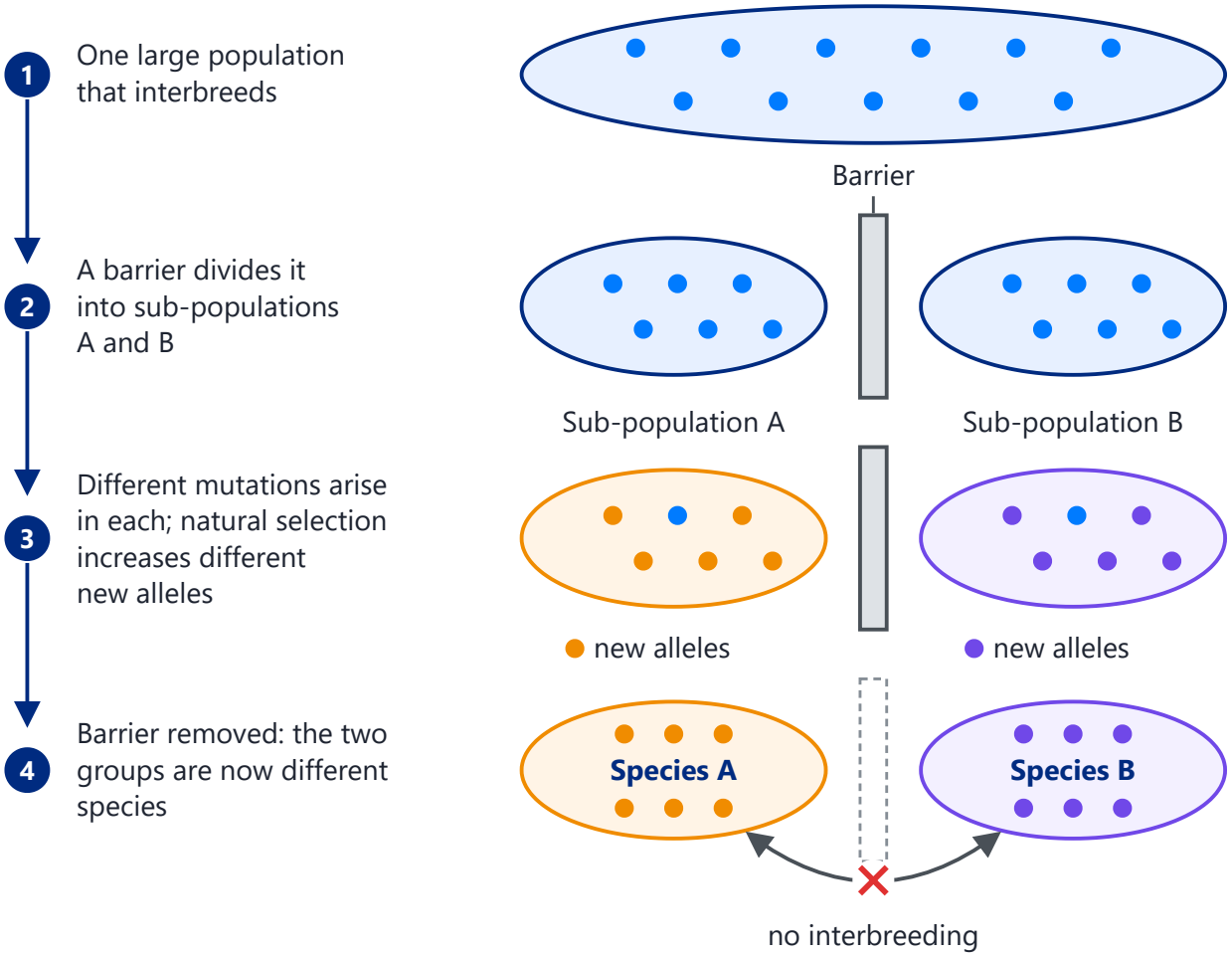
4. Heterozygotes: $2pq = 2 \times 0.7 \times 0.3 = 0.42$, so 42 % are carriers. Homozygous dominant: $p^2 = 0.49$, or 49 %.

5. Check: 0.49 + 0.42 + 0.09 = 1.00.

Reproductive isolation and new species

A **species** is a group of organisms with similar characteristics that can interbreed to produce fertile offspring. Members of a species are reproductively isolated from other species. Horses and donkeys can mate, but their offspring, mules, are sterile, so horses and donkeys remain separate species.

New species form (speciation) when populations of one species become reproductively isolated, so that genes no longer flow between them. Each population then accumulates different genetic information: different mutations arise in each, and if the populations live in different conditions, different selection pressures favour different alleles. Over many generations the populations may become so different that they could no longer interbreed to produce fertile offspring even if they met again. They are then separate species.



Geographical isolation leading to two new species.

- **Geographical isolation:** a physical barrier such as a mountain range, a river, a sea or a newly formed land bridge separates two populations. When the Isthmus of Panama formed about 3 million years ago, it separated marine populations on the Pacific and Caribbean sides, and many pairs of separate species have since evolved. Speciation after geographical separation is called allopatric speciation.
- **Ecological isolation:** two populations live in the same area but use different habitats within it, so they rarely meet.
- **Seasonal (temporal) isolation:** the populations breed or flower at different times of year, or different times of day.
- **Behavioural isolation:** courtship songs, displays or other signals change so that individuals no longer recognise each other as mates. Chiffchaffs and willow warblers look almost identical but have different songs and do not interbreed.
- **Mechanical isolation:** the reproductive organs, or in plants the structure of the flowers, no longer match, so mating or pollination fails.
- **Hybrid inviability or sterility:** mating happens, but the hybrid zygote does not develop, or the hybrid offspring are infertile.

When the isolating mechanism operates within one area, without a physical barrier (for example through behavioural or seasonal isolation), the process is called sympatric speciation. When one ancestral species gives

rise to many species, each adapted to a different niche, as happened with the finches of the Galápagos Islands, which evolved different beaks for different foods, the process is called adaptive radiation.

Classification

Classification organises the variety of life into groups based on the relationships between organisms. Organisms are grouped by their similarities and differences in phenotype (their observable features, including anatomy, physiology and behaviour) and, increasingly, in genotype (their DNA). The aim is for the groups to reflect evolutionary relationships, so that organisms grouped together share a recent common ancestor. Classification is built around the species concept, because the species is the basic unit that the larger groups are made from.

Organisms are placed in a hierarchy of groups (taxa), each containing the groups below it, from the largest to the smallest:

Taxon	Domestic cat	Human
Domain	Eukarya	Eukarya
Kingdom	Animalia	Animalia
Phylum	Chordata	Chordata
Class	Mammalia	Mammalia
Order	Carnivora	Primates
Family	Felidae	Hominidae
Genus	*Felis*	*Homo*
Species	*Felis catus*	*Homo sapiens*

Each species has a two-part scientific name, the binomial: the genus name with a capital letter, followed by the species name in lower case, both in italics (or underlined when handwritten), such as *Homo sapiens*. Because every scientist uses the same name, there is no confusion between different local or common names.

Molecular evidence

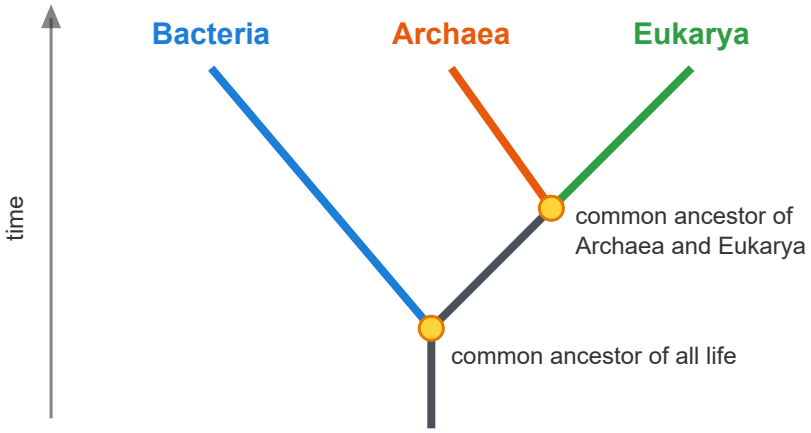
Classification used to rely almost entirely on appearance, which can mislead: unrelated organisms can look alike because they live in similar environments, and members of the same species can look very different (males and females, for example). Molecular phylogeny compares molecules instead:

- **DNA base sequences:** the more similar the sequence of bases in the same gene in two species, the more closely related they are, because fewer mutations have built up since they shared a common ancestor.
- **Amino acid sequences:** proteins found in many species, such as cytochrome c, can be compared in the same way.
- **RNA sequences:** ribosomal RNA is present in every organism and changes slowly, so it is useful for comparing very distantly related groups.

These comparisons are used to build phylogenetic trees, branching diagrams that show how species are thought to have evolved from common ancestors. Each branch point represents a common ancestor, and species that branch apart more recently are more closely related.

The three-domain system

For many years, all life was split into prokaryotes and eukaryotes, and organisms were classified into five kingdoms. From the late 1970s Carl Woese compared the base sequences of ribosomal RNA from many organisms and found that prokaryotes fall into two very different groups, as different from each other as either is from eukaryotes. In 1990 he and his colleagues proposed that life should be divided into three domains, a level of classification above the kingdom.



Feature	Bacteria	Archaea	Eukarya
Cell type	Prokaryotic	Prokaryotic	Eukaryotic
Nucleus and membrane-bound organelles such as mitochondria	Absent	Absent	Present
Peptidoglycan in the cell wall	Present	Absent	Absent
Membrane lipids	Unbranched fatty acid chains, ester-linked	Branched chains, ether-linked	Unbranched fatty acid chains, ester-linked
Histone proteins with the DNA	Absent	Present in many	Present
RNA polymerase	One type, relatively simple	One type, complex and similar to the eukaryotic enzyme	Several types, complex
First amino acid carried by tRNA at the start of a protein	Formylmethionine	Methionine	Methionine
Growth stopped by antibiotics such as streptomycin	Yes	No	No

Many archaea live in extreme environments, such as hot springs, very salty lakes or places without oxygen, although others are found in soil and the oceans. In several molecular features, such as their histones and RNA polymerases, archaea are more similar to eukaryotes than to bacteria. The domain Eukarya contains the kingdoms Protoctista, Fungi, Plantae and Animalia.

How the scientific community evaluates new ideas

Woese's proposal was not accepted at once. Like any new scientific claim, it had to be critically evaluated by other scientists before it changed how life is classified.

From a new idea to accepted science

1. The scientist publishes the findings in a scientific journal, presents them at conferences and shares data online, so that other scientists can examine them.
2. Before publication, the paper is peer reviewed: other experts in the field check that the methods are valid, the data support the conclusions and the work is original.

   Peer review filters out flawed or poorly supported work before it enters the scientific record.
3. Other scientists repeat the experiments or collect new data, for example by sequencing more organisms, to see whether the results can be reproduced.
4. If the new evidence supports the idea, it gradually gains acceptance; if not, the idea is rejected or modified. The three-domain system is now widely accepted because sequencing many more genes has supported Woese's findings.

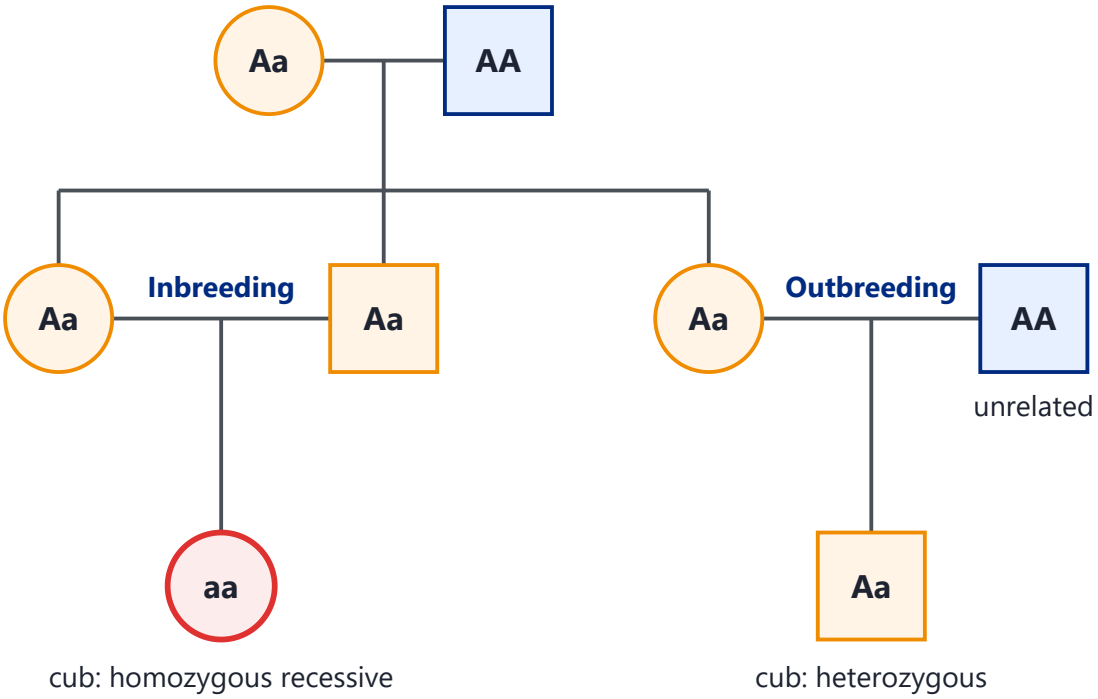
KEY IDEA

Classification is not fixed. As new evidence, especially molecular evidence, becomes available and is evaluated, organisms can be moved between groups and new groups can be created.

Threats to biodiversity

The variety of life is now being reduced faster than at any time in human history, mainly because of human activity: destruction of habitats for farming, building and logging; over-hunting and over-fishing; pollution; introduced species that compete with or prey on native species; and climate change. A species is endangered when it is at high risk of becoming extinct.

Small populations face an extra problem. With few individuals, some alleles are lost by chance in each generation, and relatives are more likely to mate with each other. This **inbreeding** increases the proportion of homozygous individuals, so harmful recessive alleles are more likely to be expressed, and the population's genetic diversity and ability to adapt both fall. **Outbreeding**, mating between unrelated individuals, increases heterozygosity and keeps genetic diversity high.



Inbreeding between two carriers can produce offspring that are homozygous for a harmful recessive allele (red). Outbreeding with an unrelated animal makes this much less likely.

Conservation: zoos and seed banks

Conservation can take place in situ, in the species' natural habitat (for example in national parks and nature reserves), or ex situ, away from the natural habitat, in zoos, botanic gardens and seed banks. Ex situ conservation is used when a population is too small, or its habitat too damaged, for it to survive in the wild.

The role of zoos

- **Captive breeding programmes:** endangered animals are bred in zoos to increase their numbers, so that the species does not become extinct. To keep genetic diversity high, zoos keep stud books (records of each animal's ancestry) and use them, together with DNA profiles, to pair unrelated animals. Animals, or their sperm for artificial insemination, are exchanged between zoos around the world, which brings new alleles into each population and avoids inbreeding. Techniques such as in vitro fertilisation and the freezing of sperm and embryos are also used.
- **Reintroduction programmes:** captive-bred animals are released into the wild to restore or reinforce wild populations. The Arabian oryx became extinct in the wild in the early 1970s; animals bred in captivity were released in Oman from 1982 and later in other countries. Release is only worthwhile if the original threat (such as hunting or habitat loss) has been removed, and animals usually need to be prepared first, for example by learning to find food and avoid predators.
- **Scientific research:** zoos study the biology, behaviour, diseases, nutrition and reproduction of endangered species, which improves breeding success in captivity and informs conservation in the wild.
- **Education:** visitors learn about endangered species, the threats they face and what people can do to help. This builds public support and funding for conservation.

Strengths of zoo programmes	Limitations
Protect animals from predators, poaching and disease, and give veterinary care	Captive populations are small, so some genetic diversity is lost even with careful breeding
Increase population size quickly, with monitored pregnancies	Some species breed poorly in captivity or need conditions that are hard to provide
Keep genetic diversity high through stud books and exchanges	Animals bred in captivity may lack the behaviour needed to survive in the wild
Provide animals for reintroduction	Reintroduction is expensive and fails if the habitat is still unsuitable or the threat remains; released animals may carry diseases to wild populations

The role of seed banks

A seed bank stores seeds in conditions that keep them alive but dormant for many years, sometimes centuries. The Millennium Seed Bank, run by the Royal Botanic Gardens, Kew, in the UK, holds seeds from many thousands of wild plant species from around the world.

How a seed bank stores seeds

1. Seeds are collected from many individual plants of a species, across its range.

   Seeds from many different parents carry a wider range of alleles, so the stored sample keeps more of the species' genetic diversity. Taking only a small fraction of the seeds does not harm the wild population.

2. The seeds are cleaned and checked, for example by X-ray, to make sure they contain a fully formed embryo and are not damaged or infested.

3. The seeds are dried to a low water content.

   With little water, metabolic reactions almost stop, so the seeds do not germinate or use up their food stores, and ice crystals that would damage cells cannot form when they are frozen.

4. They are sealed in airtight containers and stored at about −20 °C.

   Low temperature slows enzyme activity further and stops the growth of fungi and bacteria that would decompose the seeds.

5. Samples are germinated at intervals to check they are still viable. If germination falls too low, a sample is grown into adult plants to produce fresh seed for storage.

- **Advantages:** seeds are small, so huge numbers of species can be stored cheaply in a small space; they stay viable for a long time; they preserve the genetic diversity of the species; and stored seeds can be used to restore habitats, to reintroduce species, and in research, for example to find crops resistant to drought or disease.
- **Limitations:** some seeds, including those of many tropical trees such as cocoa and mango, are damaged by drying or freezing and cannot be stored this way; these are kept as growing plants in field gene banks or as tissue cultures. Seeds stored for many years do not evolve with their changing natural environment, and testing viability uses up some of the stored seed.

EXAM TIP

Plants are easier to conserve ex situ than animals: plants produce many seeds, collecting some does no harm, and dried, frozen seeds take little space and almost no care. Animals have to be kept, fed and bred as living individuals.

Quick checks

QUICK CHECK

A sample contains three species with 6, 3 and 1 individuals. What is the index of diversity, D?

- 1.25
- 2.50
- 3.00
- 0.40

N = 10, so N(N − 1) = 90. Σn(n − 1) = 30 + 6 + 0 = 36. D = 90 ÷ 36 = 2.50. Turning the fraction upside down gives 0.40, which is impossible because D can never be less than 1.

QUICK CHECK

Of 40 animals tested, 14 show two bands for a gene on a DNA gel. What is the heterozygosity index?

- 0.65
- 0.26
- 0.35
- 2.86

Animals with two bands are heterozygous, so H = 14 ÷ 40 = 0.35. The value 0.65 is the proportion of homozygotes, the commonest mix-up.

QUICK CHECK

In a population in Hardy-Weinberg equilibrium, 16 % of individuals have the recessive phenotype. What proportion are heterozygous?

- 0.16
- 0.24
- 0.36
- 0.48

$q^2 = 0.16$, so $q = 0.4$ and $p = 0.6$. Heterozygotes = $2pq = 2 \times 0.6 \times 0.4 = 0.48$. The value 0.36 is p^2, the homozygous dominant frequency. A common error is to take $q = 0.16$ instead of square-rooting it.

QUICK CHECK

Which observation best supports placing Archaea in a separate domain from Bacteria?

- Archaea have no nucleus
- The base sequences of ribosomal RNA in Archaea differ greatly from those in Bacteria
- Archaea are found only in extreme environments
- Archaea are smaller than eukaryotic cells

Woese's three-domain system was based on molecular evidence, especially rRNA sequences. Having no nucleus is shared with Bacteria, so it cannot separate them, and many archaea live in ordinary habitats.

QUICK CHECK

Which is a physiological adaptation?

- A seal's thick layer of blubber
- Birds migrating south in autumn
- A diving mammal's heart rate slowing when it dives
- The sticky hairs on a sundew leaf

Slowing the heart rate is a change in how the body's processes work, so it is physiological. Blubber and sticky hairs are structures (anatomical), and migration is behavioural.

QUICK CHECK

Why do zoos exchange animals between captive breeding programmes?

- To reduce the cost of feeding the animals
- To increase the frequency of homozygous individuals
- So that the animals become used to travelling before reintroduction
- To reduce inbreeding and bring new alleles into each population

Pairing unrelated animals from different zoos is outbreeding: it keeps heterozygosity and genetic diversity high. Increasing homozygosity is what inbreeding does, which programmes try to avoid.

QUICK CHECK

Why are seeds dried before being stored in a seed bank at −20 °C?

- Low water content slows metabolism and stops ice crystals damaging the cells when frozen
- Drying kills the embryo so that the seed cannot germinate in storage
- Drying removes the seed coat so the seeds take up less space
- Dry seeds need light to stay viable

The embryo must stay alive. Drying makes the seed dormant by slowing enzyme activity, and with little water present, damaging ice crystals cannot form during freezing.